Symplectic Beam Transport and Physics-Informed Optimization

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Abstract

We present a framework for beam-dynamics modeling and optics optimization at UH Mānoa. The document is organized in two parts. Part A develops the beam-transport foundations: kinetic continuity, Hamiltonian flow, Liouville's theorem, covariance and Twiss transport, symplectic maps, Lie algebraic transfer maps, and the relation between linear matrices and higher-order Taylor tensors. This establishes the mathematical structure used by first-order matrix models, high-order differential-algebraic maps, differentiable beam-physics surrogates, and particle tracking codes.

Part B formulates beam matching as an inverse problem in the space of controllable machine parameters. We distinguish fixed-map inversion from tunable-lattice optimization, derive response-matrix and pseudoinverse updates, and connect the singular spectrum of the Jacobian to the curvature spectrum of the objective function. The optimization landscape is treated as the pullback of the beam-transport map onto the admissible control hypercube. This motivates a layered strategy: first build a parallel topology pre-map using Sobol or other low-discrepancy samples enriched with optics, phase-advance, dispersion, Jacobian, Hessian, stability, and spectral labels; then use this structured map to initialize and constrain deterministic, Bayesian exploration, Bayesian optimization, TuRBO, population-based, or neural-surrogate methods.

The optimization framework is extended to include explicit constraint taxonomies (forbidden, probabilistic, and soft-penalty constraints), physics-informed kernels and prior means, differentiable physics coupled to Bayesian optimization, trust-region expansion and contraction, local validity radii, and solution-stability neighborhoods. The annexes collect practical material on space charge, computational scaling, coordinate conventions, linear algebra, local geometry of optimization hypersurfaces, and implementation notes for these software layers.

Contents

A Beam Transport Dynamics and Symplectic Structure 4

A.1 From Continuity to Hamiltonian Beam Dynamics 4

A.1.1 Equation of Continuity and Liouville's Theorem 4
A.1.2 Distributions, Covariances, and Closures 5
A.1.3 Hamiltonian in Electromagnetic Fields and Linear Optics 6

A.2 Hamiltonian Symplectic Maps and Lie-Transfer Formalism 6

A.2.1 Canonical phase space and the symplectic form 6
A.2.2 Hamiltonian vector fields and the Lie algebra of observables 7
A.2.3 Lie exponential representation of an element map 8
A.2.4 Composition, BCH formula, and symplectic splitting 8
A.2.5 From Hamiltonian degree to Taylor-map order 8
A.2.6 Taylor maps and transfer tensors 9

A.3 Application: Beam Transport through a Multipole 9

A.3.1 General multipole Hamiltonian 9
A.3.2 Lie-map viewpoint and order-by-order structure 11
A.3.3 Examples of nonlinear multipoles and their tensor content 11
A.3.4 Linear transport from the quadratic Hamiltonian 12
A.3.5 Example: hard-edge quadrupole 12
A.3.6 Example: sector dipole 13
A.3.7 Drift as a limiting case 14

Annex A: Space Charge and Impedance 15

Annex B: Computational Cost and Error Scaling 15

Annex C: Coordinates, Laplacians, and Conventions 15

Annex D (Draft): Thermionic Emission Model 16

References 17

B Beam Transport Optimization 18

B.1 Inverse Optics and Local Objective Geometry 18

B.1.1 Inverse use of a fixed linear transport map 19
B.1.2 Matching as an inverse problem: response matrices and regularization . . . 20
B.1.3 Objective design, normalization, and transformed losses 23
B.1.4 Objective-function curvature: Jacobian, Hessian, and magnet modes 24
B.1.5 Computational cost of Jacobian and Hessian information 26
B.1.6 Phase advance, symplectic eigenvalues, and normal-form coordinates 26
B.1.7 High-order maps and consistency of the response 28
B.1.8 Response-based splitting of the beamline into sections 28

B.2 Topology Mapping of the Matching Landscape 29

B.2.1 Basins, islands, and the pullback objective 29
B.2.2 Parallel Sobol pre-mapping with optics and curvature labels 30
B.2.3 Graph descent and basin classification in augmented feature space 30

B.2.4 Targeted Fourier and frequency analysis along physics directions 30
B.2.5 Neural-network layer on top of the physics classifiers 32

B.3 Optimization and the UH Workflow 32
B.3.1 Deterministic optimization algorithms 32
B.3.1.1 Derivative-free local methods: Nelder–Mead and Powell 33
B.3.1.2 Finite-difference and quasi-Newton methods: BFGS and L-BFGS-B 33
B.3.1.3 Newton and trust-region methods 33
B.3.1.4 Constrained local methods: SLSQP, COBYLA, and COBYQA . . 34
B.3.1.5 Role in the UHM benchmark 34
B.3.2 Bayesian optimization and exploration 34
B.3.2.1 Exploration and exploitation 35
B.3.2.2 Surrogate models 35
B.3.2.3 Kernel design 36
B.3.2.4 Prior means 38
B.3.3 Constraints and feasibility 38
B.3.3.1 Forbidden conditions 38
B.3.3.2 Probabilistic constraints 38
B.3.3.3 Soft penalties 39
B.3.3.4 Feasibility-aware acquisition 39
B.3.4 Local Bayesian optimization 39
B.3.4.1 Trust regions 39
B.3.4.2 Region adaptation 40
B.3.4.3 Multiple regions 40
B.3.4.4 Validity radius 40
B.3.5 Robust solutions 40
B.3.5.1 Parameter sensitivity 40
B.3.5.2 Feasible neighborhoods 41
B.3.5.3 Solution families 41
B.3.6 UH workflow 41
B.3.6.1 Topology-informed initialization 41
B.3.6.2 Adaptive sequence 41
B.3.6.3 Problem regimes 43

Annex A: Linear Algebra for Response Maps 43

Annex B: Local Geometry of the Optimization Landscape 44

Annex C: Implementation Notes for Optimization and BO 44

References 47

Part A

Beam Transport Dynamics and Symplectic Structure

Scope of Part A. This part establishes the beam-transport model used later by the optimization framework. We start from kinetic continuity and the conditions under which the beam evolution reduces to Hamiltonian flow and Liouville transport. We then pass to low-order moment descriptions, Courant–Snyder parameters, covariance propagation, and finally to symplectic maps expressed through Lie operators and Taylor tensors. The aim is to keep the forward model general enough to cover first-order matrix optics, high-order differential-algebraic maps, differentiable beam-dynamics models, and particle-based tracking, while retaining the Hamiltonian structure whenever the underlying physics permits it.

The notation and physical interpretation follow the standard linear-optics and accelerator-physics literature [1, 2, 3], the Lie-map and symplectic-map formulation of nonlinear beam dynamics [4, 5], and the moment-closure ideas used in rms beam descriptions [6, 7]. Kinetic and non-Hamiltonian extensions are included only to identify where the ideal symplectic picture must be modified, for example by sources, losses, radiation, collective effects, or dissipative wake models [8]. In Part B, these same structures reappear as optimization features: phase advances, symplectic eigenstructure, response modes, and higher-order transport information become labels for topology mapping, priors for surrogate models, correlation structures for Bayesian optimization, and metrics for anisotropic trust regions.

A.1 From Continuity to Hamiltonian Beam Dynamics

A.1.1 Equation of Continuity and Liouville's Theorem

Let $f(t, \mathbf{x}, \mathbf{p}) \geq 0$ be the one-particle distribution function in phase space $\mathbf{z} = (\mathbf{x}, \mathbf{p}) \in \mathbb{R}^6$. The most general kinetic balance we consider is

$$\partial_t f + \mathbf{v} \cdot \nabla_{\mathbf{x}} f + \mathbf{F} \cdot \nabla_{\mathbf{p}} f = Q[f] + S[f], \qquad \mathbf{v} = \frac{\mathbf{p}}{\gamma m},\ \gamma = \sqrt{1 + \frac{\mathbf{p}^2}{m^2 c^2}}, \tag{A.1.1}$$

with forces $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$ (external and/or self-consistent), collision operator $Q[f]$ (Boltzmann/Landau/Fokker–Planck), and source/sink $S[f]$ (e.g. radiation, apertures, injection).

Reduction to Hamiltonian continuity. If $Q[f] \equiv S[f] \equiv 0$ and $(\mathbf{E}, \mathbf{B})$ derive from a Hamiltonian $H(\mathbf{z}, t)$ (see §A.1.3), then the characteristics obey Hamilton's equations

$$\dot{\mathbf{z}} = \mathbf{J}\, \nabla_{\mathbf{z}} H(\mathbf{z}, t), \qquad \mathbf{J} = \begin{pmatrix} 0 & I_3 \\ -I_3 & 0 \end{pmatrix}, \tag{A.1.2}$$

and the continuity equation (A.1.1) becomes

$$\partial_t f + \nabla_{\mathbf{z}} \cdot \big(f\, \dot{\mathbf{z}}\big) = \partial_t f + \nabla_{\mathbf{z}} \cdot \big(f\, \mathbf{J} \nabla_{\mathbf{z}} H\big) = 0. \tag{A.1.3}$$

Using $\nabla \cdot (\mathbf{J} \nabla H) = 0$ (skew-symmetry of $\mathbf{J}$), (A.1.3) is equivalent to the *Liouville equation*

$$\frac{Df}{Dt} \equiv \partial_t f + \{f, H\} = 0, \qquad \{F, G\} := \nabla F^{\top} \mathbf{J} \nabla G, \tag{A.1.4}$$

which states that f is constant along Hamiltonian trajectories (phase-space volume preservation). Thus:

$$\boxed{\text{Hamiltonian flow} \iff \text{Liouville's theorem for } f.}$$

When Q or S are nonzero, or when forces are not derived from a Hamiltonian, (A.1.4) no longer holds and phase-space volume need not be preserved.

In the kinetic description above, $\mathbf{p}$ denotes the mechanical momentum. In the Hamiltonian beam-optics formulation used below, we switch to canonical phase-space coordinates adapted to the reference trajectory.

A.1.2 Distributions, Covariances, and Closures

In accelerator modeling we often represent the beam either by samples from f (particles) or by its low-order moments. The *beam covariance* (or Σ-matrix) in 6D is

$$\Sigma = \mathbb{E}\big[(\mathbf{z}-\bar{\mathbf{z}})(\mathbf{z}-\bar{\mathbf{z}})^\top\big] \in \mathbb{R}^{6\times 6}, \quad \Sigma = \Sigma^\top \succeq 0, \tag{A.1.5}$$

with 15 independent second moments. In a 2D canonical plane (u, p_u),

$$\Sigma_{(u)} = \begin{pmatrix} \langle u^2\rangle & \langle u p_u\rangle \\ \langle u p_u\rangle & \langle p_u^2\rangle \end{pmatrix} = \varepsilon_u \begin{pmatrix} \beta_u & -\alpha_u \\ -\alpha_u & \gamma_u \end{pmatrix}, \qquad \beta_u\gamma_u - \alpha_u^2 = 1, \tag{A.1.6}$$

so that the Courant–Snyder ellipse $\gamma_u u^2 + 2\alpha_u u p_u + \beta_u p_u^2 = \varepsilon_u$ has area $\pi\varepsilon_u$; the envelope is $R_u = \sqrt{\varepsilon_u\beta_u}$.

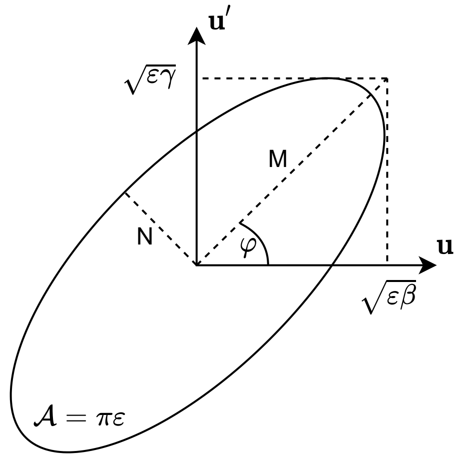

Figure 1. Phase-space ellipse (2D) with Twiss parameters. Extend to 6D by considering subspace projections and cross-correlations in Σ.

Closure Moment transport under nonlinear maps introduces dependencies on higher moments (third, fourth, ...). A *closure* expresses these higher moments as functionals of the lower ones (e.g. of Σ), thereby closing the system. Examples:

- **Gaussian (Wick/Isserlis).** Odd central moments vanish; even moments factor into sums of products of Σ entries. This exactly closes fourth and sixth moments in terms of Σ.
- **KV (Kapchinskij–Vladimirskij)/Waterbag.** Uniform on an ellipse/finite region. Odd moments vanish by symmetry; even moments do *not* obey Wick and must be computed by geometry-specific integrals. KV yields exact RMS envelope equations for linear external + linear self-field.
- **Gauss–Hermite.** Expand $f = \phi_\Sigma(\mathbf{z})\big[1+\sum_{|\alpha|\geq 3} c_\alpha H_\alpha(\mathbf{z})\big]$; orthogonality gives controlled corrections to Gaussian closure (skew/kurtosis).

When closures are valid, covariance transport can replace particle transport at far lower cost (§B.1).

A.1.3 Hamiltonian in Electromagnetic Fields and Linear Optics

With potentials $(\Phi, \mathbf{A})$, the relativistic single-particle Hamiltonian is

$$H(\mathbf{x}, \mathbf{p}; t) = \sqrt{m^2c^4 + c^2\big(\mathbf{p} - q\mathbf{A}(\mathbf{x}, t)\big)^2} \ + \ q\,\Phi(\mathbf{x}, t). \tag{A.1.7}$$

Using s (path length) as independent variable and expanding around a reference trajectory of momentum p_0, we write

$$H(\mathbf{z}; s) = H^{(2)}(\mathbf{z}; s) \ + \ H^{(3)}(\mathbf{z}; s) \ + \ H^{(4)}(\mathbf{z}; s) + \cdots, \tag{A.1.8}$$

where $H^{(2)}$ produces *linear* optics (drifts, quads, sector dipoles with dispersion) and $H^{(3)}, H^{(4)}$ generate sextupolar, octupolar, fringe and chromatic effects.

Linearization gives Hill's equations,

$$u''(s) + K_u(s)\,u(s) = 0, \qquad u \in \{x, y\}, \tag{A.1.9}$$

with periodic focusing $K_u(s + L) = K_u(s)$. Floquet theory yields $u = \sqrt{\beta_u(s)}\,A\cos(\psi_u(s) + \phi)$ and the exact identity

$$\frac{1}{2}\beta_u\beta_u'' - \frac{(\beta_u')^2}{4} + K_u(s)\,\beta_u^2 = 1. \tag{A.1.10}$$

Relativistically, focusing scales as $(\beta\gamma)^{-1}$ and direct space-charge forces weaken as γ^{-2} in the beam frame; normalized emittance $\varepsilon_{n,u} = \beta\gamma\,\varepsilon_u$ is invariant for Hamiltonian acceleration.

A.2 Hamiltonian Symplectic Maps and Lie-Transfer Formalism

The previous sections introduced Hamiltonian flow and the covariance description of a beam. We now collect the mathematical structure that makes accelerator transfer maps special. The aim is not only to state the symplectic condition, but to connect four equivalent viewpoints used in beam dynamics:

(i) canonical Hamiltonian flow in phase space;

(ii) symplectic maps and matrices;

(iii) Lie operators generated by Hamiltonians;

(iv) Taylor maps and high-order transfer tensors.

This structure is central for Part B: the optimization objective is not an arbitrary black-box function of machine parameters. It is induced by a symplectic, or approximately symplectic, beam-transport map whose eigenvalues, phase advances, normal modes, and response tensors shape the matching landscape. These quantities later enter topology features, prior means, kernel correlations, and validity metrics for the optimization layer.

A.2.1 Canonical phase space and the symplectic form

Let

$$\mathbf{z} = (q_1, p_1, q_2, p_2, q_3, p_3)^\top \in \mathbb{R}^6 \tag{A.2.1}$$

denote canonical phase-space coordinates and let

$$\mathbf{J} = \begin{pmatrix} 0 & I_3 \\ -I_3 & 0 \end{pmatrix}, \qquad \mathbf{J}^\top = -\mathbf{J}, \qquad \mathbf{J}^{-1} = -\mathbf{J}. \tag{A.2.2}$$

The canonical two-form is

$$\omega(\delta\mathbf{z}_1, \delta\mathbf{z}_2) = \delta\mathbf{z}_1^\top \mathbf{J}\delta\mathbf{z}_2. \tag{A.2.3}$$

It measures oriented phase-space area in each canonical plane and generalizes the Courant–Snyder ellipse area to coupled multidimensional motion.

Definition A.2.1 (Symplectic map)**.** *A differentiable map* $\mathcal{M} : \mathbb{R}^{2d} \to \mathbb{R}^{2d}$ *is symplectic if its Jacobian preserves the two-form,*

$$D\mathcal{M}(\mathbf{z})^\top \mathbf{J} D\mathcal{M}(\mathbf{z}) = \mathbf{J}. \tag{A.2.4}$$

For a linear map $\mathbf{z}_{\text{out}} = R\mathbf{z}_{\text{in}}$*, this reduces to*

$$R^\top \mathbf{J} R = \mathbf{J}. \tag{A.2.5}$$

Proposition A.2.1 (Basic properties of symplectic matrices)**.** *If* $R \in Sp(2d, \mathbb{R})$*, then:*

$$R^{-1} = -\mathbf{J} R^\top \mathbf{J}, \tag{A.2.6}$$

$$\det R = 1, \tag{A.2.7}$$

$$\lambda \in \operatorname{spec}(R) \Longrightarrow \lambda^{-1}, \lambda^*, (\lambda^*)^{-1} \in \operatorname{spec}(R) \quad \textit{whenever the corresponding values are distinct.} \tag{A.2.8}$$

Thus stable uncoupled motion has eigenvalues on the unit circle, $\lambda = e^{\pm i\mu}$*, where* μ *is a phase advance.*

Remark A.2.1 (Liouville theorem versus symplecticity)**.** *Liouville's theorem states preservation of phase-space volume. Symplecticity is stronger: it preserves the two-form and therefore all canonical phase-plane areas and their coupled generalizations. A Hamiltonian flow is symplectic and therefore volume preserving. A dissipative effect, aperture loss, stochastic emission, or non-Hamiltonian radiation model may preserve neither exactly.*

A.2.2 Hamiltonian vector fields and the Lie algebra of observables

For a Hamiltonian $H(\mathbf{z}; s)$, Hamilton's equations may be written

$$\frac{d\mathbf{z}}{ds} = \mathbf{J}\nabla_{\mathbf{z}} H(\mathbf{z}; s). \tag{A.2.9}$$

Every phase-space function $F(\mathbf{z})$ evolves according to

$$\frac{dF}{ds} = \{F, H\}, \qquad \{F, G\} = \nabla F^\top \mathbf{J} \nabla G. \tag{A.2.10}$$

The Poisson bracket is bilinear, antisymmetric, satisfies the Leibniz rule, and obeys the Jacobi identity,

$$\{F, \{G, H\}\} + \{G, \{H, F\}\} + \{H, \{F, G\}\} = 0. \tag{A.2.11}$$

Therefore smooth phase-space functions form a Lie algebra modulo constants.

Definition A.2.2 (Lie operator)**.** *For a smooth function* F*, the Lie operator* $:F:$ *is defined by*

$$:F:\, G = \{G, F\}. \tag{A.2.12}$$

The commutator of Lie operators is again a Lie operator,

$$[:F:, :G:] =: \{F, G\} :, \tag{A.2.13}$$

up to the sign convention used for the Poisson bracket.

Lemma A.2.1 (Hamiltonian flows are symplectic)**.** *Let* φ_s *be the flow of Eq.* (A.2.9)*. Then*

$$D\varphi_s(\mathbf{z})^\top \mathbf{J} D\varphi_s(\mathbf{z}) = \mathbf{J}. \tag{A.2.14}$$

Consequently, Hamiltonian transfer maps preserve the canonical two-form and satisfy Liouville's theorem.

A.2.3 Lie exponential representation of an element map

If H is independent of s over a step Δs, the exact Hamiltonian map may be written formally as

$$\mathbf{z}_{\text{out}} = \exp\big(:H:\ \Delta s\big)\, \mathbf{z}_{\text{in}}. \tag{A.2.15}$$

Applying the exponential to one coordinate gives the Lie series

$$z_i^{\text{out}} = e^{:H:\Delta s} z_i^{\text{in}} = z_i^{\text{in}} + \Delta s\ :H:\, z_i^{\text{in}} + \frac{\Delta s^2}{2!}\ :H:^2\, z_i^{\text{in}} + \frac{\Delta s^3}{3!}\ :H:^3\, z_i^{\text{in}} + \cdots . \tag{A.2.16}$$

The first term is the infinitesimal Hamiltonian kick; higher terms represent repeated Poisson-bracket actions. Thus the Lie exponential is not an approximation by itself. It is the exact formal map generated by the Hamiltonian; approximations enter only when the exponential is truncated, factorized, or evaluated using a finite-order map.

A.2.4 Composition, BCH formula, and symplectic splitting

If $H = H_1 + H_2$, the exact flow is generally not the product of the two exact subflows. The Baker–Campbell–Hausdorff (BCH) formula gives

$$e^{:H_1:}e^{:H_2:} = e^{:H_1+H_2+\frac{1}{2}[H_1,H_2]+\frac{1}{12}\big([H_1,[H_1,H_2]]+[H_2,[H_2,H_1]]\big)+\cdots:}, \tag{A.2.17}$$

$$[H_1, H_2] := \{H_1, H_2\}. \tag{A.2.18}$$

The commutator terms quantify the error made when one composes separate drifts, kicks, rf gaps, space-charge kicks, or radiation approximations. Symmetric factorizations, such as Forest–Ruth or Yoshida splittings, cancel the lowest commutator errors while preserving symplecticity by construction [4, 5]. This is why accelerator tracking codes often prefer split maps over generic ODE integrators: the numerical map remains close to the correct Hamiltonian geometry over long transport distances or many turns.

A.2.5 From Hamiltonian degree to Taylor-map order

The Lie form and the Taylor-map form are two representations of the same transformation. Let the Hamiltonian be expanded by homogeneous degree in the canonical variables,

$$H(\mathbf{z}; s) = H^{(2)}(\mathbf{z}; s) + H^{(3)}(\mathbf{z}; s) + H^{(4)}(\mathbf{z}; s) + \cdots . \tag{A.2.19}$$

The part $H^{(m)}$ homogeneous of degree m contributes a polynomial of degree $m-1$ to the first-order Lie action on a coordinate. Indeed,

$$\{z_i, H^{(m)}\} = \sum_{j=1}^{3}\left(\frac{\partial z_i}{\partial q_j}\frac{\partial H^{(m)}}{\partial p_j} - \frac{\partial z_i}{\partial p_j}\frac{\partial H^{(m)}}{\partial q_j}\right), \tag{A.2.20}$$

and differentiating a homogeneous polynomial of degree m lowers its degree by one:

$$\deg\{z_i, H^{(m)}\} = m - 1. \tag{A.2.21}$$

Thus

$$H^{(2)} \Rightarrow \text{linear map}, \qquad H^{(3)} \Rightarrow \text{quadratic map terms}, \qquad H^{(4)} \Rightarrow \text{cubic map terms}. \tag{A.2.22}$$

The quadratic Hamiltonian generates the first-order matrix R, the cubic Hamiltonian generates the second-order tensor T, and the quartic Hamiltonian generates the third-order tensor U.

A.2.6 Taylor maps and transfer tensors

Let

$$\mathbf{z} = (x, p_x, y, p_y, \zeta, \delta)^\top \tag{A.2.23}$$

be canonical coordinates. A smooth beamline map F expanded about the reference input $\mathbf{z}_{\text{in}} = 0$ has components

$$z_i^{\text{out}} = a_i + \sum_j R_{ij} z_j^{\text{in}} + \frac{1}{2!} \sum_{j,k} T_{ijk} z_j^{\text{in}} z_k^{\text{in}} + \frac{1}{3!} \sum_{j,k,l} U_{ijkl} z_j^{\text{in}} z_k^{\text{in}} z_l^{\text{in}} + \cdots, \tag{A.2.24}$$

where

$$a_i = F_i(0), \tag{A.2.25}$$

$$R_{ij} = \left. \frac{\partial F_i}{\partial z_j} \right|_0, \tag{A.2.26}$$

$$T_{ijk} = \left. \frac{\partial^2 F_i}{\partial z_j \partial z_k} \right|_0 = T_{ikj}, \tag{A.2.27}$$

$$U_{ijkl} = \left. \frac{\partial^2 F_i}{\partial z_j \partial z_k} z_l \right|_0, \tag{A.2.28}$$

and similarly for higher orders. The constant term a_i represents a reference-orbit offset, steering kick, misalignment, or expansion about an orbit that is not exactly the design trajectory. In an ideal centered lattice one often has $a_i = 0$.

Remark A.2.2 (Symplectic constraints on Taylor tensors)**.** *A finite Taylor expansion with arbitrary coefficients is not automatically symplectic. Symplecticity imposes algebraic constraints coupling $R, T, U, \ldots$. Lie-map and differential-algebraic methods are useful because they generate these tensors from Hamiltonian generators, preserving the underlying symplectic structure order by order [4, 5].*

A.3 Application: Beam Transport through a Multipole

This appendix complements the discussion of Hamiltonian splitting, Lie maps, and Taylor maps given in §A.1.3 and §A.2. Rather than repeating the general definitions of the Taylor coefficients a_i, R_{ij}, $T_{i\,jk}$, and $U_{i\,jkl}$, we start here from a generic multipole Hamiltonian and show how the first-, second-, and third-order map coefficients arise from the order-by-order expansion of the Hamiltonian. We then specialize to the first-order transport of the quadrupole and the sector dipole.

We use canonical coordinates

$$\mathbf{z} = (x, p_x, y, p_y, \zeta, \delta)^\top,$$

with independent variable s along the reference trajectory. Longitudinal conventions may vary, provided the normalization remains canonical.

A.3.1 General multipole Hamiltonian

In a magnetic lattice, after choosing a reference orbit and expanding about it, the beam dynamics may be described by a Hamiltonian of the form

$$H(\mathbf{z}; s) = H_0(\delta; s) + H_1(\mathbf{z}; s) + H_2(\mathbf{z}; s) + H_3(\mathbf{z}; s) + H_4(\mathbf{z}; s) + \cdots, \tag{A.3.1}$$

where H_n denotes the part homogeneous of degree n in the canonical variables. The constant term H_0 does not affect the equations of motion, and H_1 vanishes when the reference trajectory is chosen consistently. The quadratic term H_2 generates linear transport, while H_3, H_4, ... generate successively higher-order nonlinear corrections.

A convenient representation of a normal magnetic multipole uses the complex transverse potential

$$\Phi_m(x, y; s) = \mathrm{Re}\left[\frac{C_m(s)}{m!}(x + iy)^m\right], \tag{A.3.2}$$

where $m = 1, 2, 3, \ldots$ corresponds respectively to dipole, quadrupole, sextupole, octupole, and higher multipoles, and $C_m(s)$ is the corresponding strength function. Expanding (A.3.2) gives the familiar normal-multipole polynomials:

$$m = 1: \qquad \Phi_1 \propto x, \tag{A.3.3}$$

$$m = 2: \qquad \Phi_2 \propto x^2 - y^2, \tag{A.3.4}$$

$$m = 3: \qquad \Phi_3 \propto x^3 - 3xy^2, \tag{A.3.5}$$

$$m = 4: \qquad \Phi_4 \propto x^4 - 6x^2y^2 + y^4. \tag{A.3.6}$$

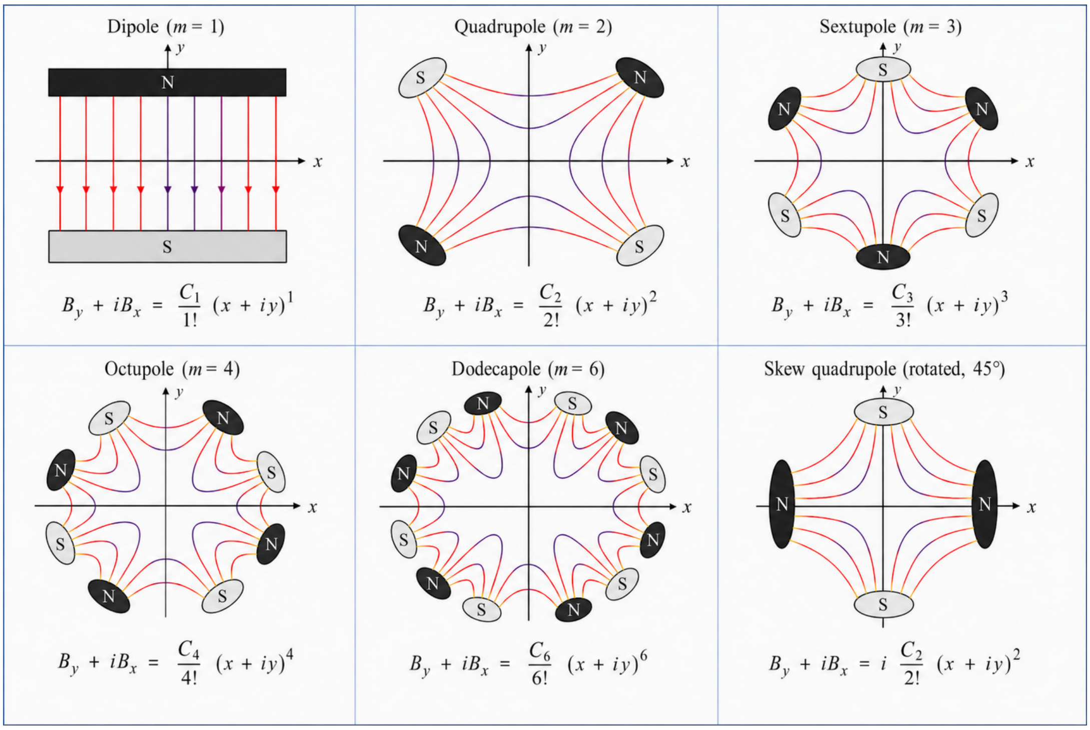

Figure 2. Multipoles and skew quadrupole configurations of magnetic field lines.

At the Hamiltonian level, the corresponding element contributes polynomial terms whose degree in the phase-space variables determines the order of the resulting map. Symbolically, one may write

$$H = H^{(2)} + H^{(3)} + H^{(4)} + \cdots, \tag{A.3.7}$$

where:

- $H^{(2)}$ contains the kinetic terms, reference-curvature terms, and linear focusing, and generates the first-order matrix R;
- $H^{(3)}$ contains the cubic terms, for example sextupolar or chromatic contributions, and generates the second-order tensor T;
- $H^{(4)}$ contains quartic terms, for example octupolar contributions, and generates the third-order tensor U.

Thus, in the notation of Eq. (A.2.24), one has schematically

$$H^{(2)} \Longrightarrow R, \qquad H^{(3)} \Longrightarrow T, \qquad H^{(4)} \Longrightarrow U, \tag{A.3.8}$$

with higher Hamiltonian orders contributing to higher Taylor tensors.

A.3.2 Lie-map viewpoint and order-by-order structure

As discussed in §A.2, the map of an element over a distance Δs is generated by the Lie exponential

$$\mathcal{M}_{\Delta s} = \exp\big(:H: \Delta s\big). \tag{A.3.9}$$

Using the decomposition (A.3.7), one may formally write

$$\mathcal{M}_{\Delta s} = \exp\big(:H^{(2)}: \Delta s\big)\exp\big(:H^{(3)}: \Delta s\big)\exp\big(:H^{(4)}: \Delta s\big)\cdots, \tag{A.3.10}$$

with Baker–Campbell–Hausdorff corrections as in Eq. (A.2.17). Expanding the resulting map in the canonical coordinates gives precisely the Taylor form already introduced in Eq. (A.2.24). We therefore do not repeat the full Taylor expansion here.

The important structural point is that truncating the Hamiltonian at quadratic order yields a purely linear map, while retaining cubic and quartic terms yields the tensors T and U of the nonlinear map. In particular:

- if $H \approx H^{(2)}$, then only the matrix R is nonzero;
- if $H \approx H^{(2)} + H^{(3)}$, then the map contains both R and T;
- if $H \approx H^{(2)} + H^{(3)} + H^{(4)}$, then the map contains R, T, and U.

A.3.3 Examples of nonlinear multipoles and their tensor content

This structure is particularly clear for sextupoles and octupoles. A normal sextupole adds a cubic Hamiltonian term of the form

$$H^{(3)}_{\text{sext}} \propto x^3 - 3xy^2, \tag{A.3.11}$$

while a normal octupole adds a quartic term

$$H^{(4)}_{\text{oct}} \propto x^4 - 6x^2y^2 + y^4. \tag{A.3.12}$$

In the thin-lens approximation, the associated canonical kicks in transverse momenta are

$$\text{(sextupole)} \qquad \Delta p_x = -m(x^2 - y^2), \qquad \Delta p_y = +2m\,xy, \tag{A.3.13}$$

and

$$\text{(octupole)} \qquad \Delta p_x = -o(x^3 - 3xy^2), \qquad \Delta p_y = -o(y^3 - 3x^2y), \tag{A.3.14}$$

where m and o denote the integrated sextupole and octupole strengths, respectively.

These expressions make transparent the connection with the Taylor tensors:

- the sextupole kick is quadratic in (x, y), hence it contributes to the second-order Taylor tensor T;
- the octupole kick is cubic in (x, y), hence it contributes to the third-order Taylor tensor U.

Chromatic effects appear when the Hamiltonian depends explicitly on δ, and fringe-field or hard-edge corrections may introduce additional cubic or quartic terms. In that case, the same logic applies: each polynomial order in the Hamiltonian feeds the corresponding order of the Taylor map.

A.3.4 Linear transport from the quadratic Hamiltonian

When the Hamiltonian is truncated at second order,

$$H \approx H^{(2)},$$

Hamilton's equations are linear in the canonical variables. In vector form one may write

$$\frac{\mathrm{d}\mathbf{z}}{\mathrm{d}s} = A(s)\,\mathbf{z}, \tag{A.3.15}$$

where the matrix $A(s)$ is determined by the quadratic Hamiltonian. The first-order transfer matrix $R(s_f; s_i)$ is then the fundamental solution of

$$\frac{\mathrm{d}R}{\mathrm{d}s} = A(s)\,R, \qquad R(s_i; s_i) = I, \tag{A.3.16}$$

so that

$$\mathbf{z}(s_f) = R(s_f; s_i)\,\mathbf{z}(s_i). \tag{A.3.17}$$

Because the system is Hamiltonian, R is symplectic.

A.3.5 Example: hard-edge quadrupole

For an ideal normal quadrupole of constant strength k and length L, the quadratic Hamiltonian in the transverse degrees of freedom is

$$H_2^{(\mathrm{quad})} = \frac{1}{2}p_x^2 + \frac{1}{2}k\,x^2 + \frac{1}{2}p_y^2 - \frac{1}{2}k\,y^2. \tag{A.3.18}$$

The corresponding Hamilton equations are

$$x' = \frac{\partial H_2^{(\mathrm{quad})}}{\partial p_x} = p_x, \qquad p_x' = -\frac{\partial H_2^{(\mathrm{quad})}}{\partial x} = -kx, \tag{A.3.19}$$

$$y' = \frac{\partial H_2^{(\mathrm{quad})}}{\partial p_y} = p_y, \qquad p_y' = -\frac{\partial H_2^{(\mathrm{quad})}}{\partial y} = +ky. \tag{A.3.20}$$

Differentiating once more and eliminating the canonical momenta gives the second-order equations

$$x'' + kx = 0, \qquad y'' - ky = 0. \tag{A.3.21}$$

For $k > 0$, the horizontal equation is that of a harmonic oscillator, so its general solution is

$$x(s) = A\cos\left(\sqrt{k}\,s\right) + B\sin\left(\sqrt{k}\,s\right), \tag{A.3.22}$$

with

$$p_x(s) = x'(s) = -A\sqrt{k}\sin\left(\sqrt{k}\,s\right) + B\sqrt{k}\cos\left(\sqrt{k}\,s\right). \tag{A.3.23}$$

Imposing the initial conditions at the quadrupole entrance, $x(0) = x_0$ and $p_x(0) = p_{x0}$, one finds

$$A = x_0, \qquad B = \frac{p_{x0}}{\sqrt{k}},$$

so that at the exit $s = L$,

$$x(L) = x_0\cos\left(\sqrt{k}L\right) + \frac{\sin\left(\sqrt{k}L\right)}{\sqrt{k}}\,p_{x0}, \tag{A.3.24}$$

$$p_x(L) = -\sqrt{k}\sin\left(\sqrt{k}L\right)x_0 + \cos\left(\sqrt{k}L\right)p_{x0}. \tag{A.3.25}$$

This immediately gives the horizontal first-order transport matrix

$$R_x^{(\mathrm{quad})} = \begin{pmatrix} \cos\left(\sqrt{k}L\right) & \dfrac{\sin\left(\sqrt{k}L\right)}{\sqrt{k}} \\ -\sqrt{k}\sin\left(\sqrt{k}L\right) & \cos\left(\sqrt{k}L\right) \end{pmatrix}. \tag{A.3.26}$$

In the vertical plane, for $k > 0$, the equation $y'' - ky = 0$ has exponential solutions, which are more conveniently written in hyperbolic form:

$$y(s) = C\cosh\left(\sqrt{k}\,s\right) + D\sinh\left(\sqrt{k}\,s\right), \tag{A.3.27}$$

with

$$p_y(s) = y'(s) = C\sqrt{k}\sinh\left(\sqrt{k}\,s\right) + D\sqrt{k}\cosh\left(\sqrt{k}\,s\right). \tag{A.3.28}$$

Using the initial conditions $y(0) = y_0$ and $p_y(0) = p_{y0}$ gives

$$C = y_0, \qquad D = \frac{p_{y0}}{\sqrt{k}},$$

hence

$$y(L) = y_0\cosh\left(\sqrt{k}L\right) + \frac{\sinh\left(\sqrt{k}L\right)}{\sqrt{k}}\,p_{y0}, \tag{A.3.29}$$

$$p_y(L) = \sqrt{k}\sinh\left(\sqrt{k}L\right)y_0 + \cosh\left(\sqrt{k}L\right)p_{y0}. \tag{A.3.30}$$

The vertical first-order transport matrix is therefore

$$R_y^{(\mathrm{quad})} = \begin{pmatrix} \cosh\left(\sqrt{k}L\right) & \dfrac{\sinh\left(\sqrt{k}L\right)}{\sqrt{k}} \\ \sqrt{k}\sinh\left(\sqrt{k}L\right) & \cosh\left(\sqrt{k}L\right) \end{pmatrix}. \tag{A.3.31}$$

If $k < 0$, the roles of the trigonometric and hyperbolic functions are exchanged between the two transverse planes. The quadrupole then becomes focusing in the y-plane and defocusing in the x-plane. The full 6×6 matrix is obtained by embedding these blocks into the chosen global coordinate system.

A.3.6 Example: sector dipole

For a sector dipole of constant curvature $h = 1/\rho$ and length L, with bend angle $\theta = hL$, the quadratic Hamiltonian about the reference orbit yields horizontal focusing together with dispersion. To first order, the horizontal dynamics obey

$$x'' + h^2 x = h\,\delta, \tag{A.3.32}$$

while the vertical plane is uncoupled and, in the simplest model without edge focusing, satisfies

$$y'' = 0. \tag{A.3.33}$$

Equation (A.3.32) is a linear inhomogeneous second-order differential equation. Its homogeneous part,

$$x'' + h^2 x = 0,$$

has the oscillatory solution

$$x_h(s) = A\cos(hs) + B\sin(hs), \tag{A.3.34}$$

while a particular solution of the full equation is obtained by taking $x_p = \delta/h$, since δ is constant to first order in this model. The general solution is therefore

$$x(s) = A\cos(hs) + B\sin(hs) + \frac{\delta}{h}. \tag{A.3.35}$$

Differentiating with respect to s gives

$$p_x(s) = x'(s) = -Ah\sin(hs) + Bh\cos(hs). \tag{A.3.36}$$

Applying the entrance conditions $x(0) = x_0$ and $p_x(0) = p_{x0}$ yields

$$A = x_0 - \frac{\delta}{h}, \qquad B = \frac{p_{x0}}{h}.$$

Substituting these constants into (A.3.35) and (A.3.36), and evaluating at $s = L$, gives

$$x(L) = x_0\cos(hL) + \frac{\sin(hL)}{h}\,p_{x0} + \frac{\delta}{h}\big(1-\cos(hL)\big), \tag{A.3.37}$$

$$p_x(L) = -h\sin(hL)\,x_0 + \cos(hL)\,p_{x0} + \delta\,\sin(hL). \tag{A.3.38}$$

Using $\rho = 1/h$ and $\theta = hL$, one obtains the standard first-order sector-dipole map

$$\begin{pmatrix} x \\ p_x \end{pmatrix}_{\text{out}} = \begin{pmatrix} \cos\theta & \rho\sin\theta \\ -\dfrac{\sin\theta}{\rho} & \cos\theta \end{pmatrix} \begin{pmatrix} x \\ p_x \end{pmatrix}_{\text{in}} + \begin{pmatrix} \rho(1-\cos\theta) \\ \sin\theta \end{pmatrix} \delta_{\text{in}}. \tag{A.3.39}$$

Equivalently, in the subspace (x, p_x, δ),

$$R^{(\text{dip})}_{x,\delta} = \begin{pmatrix} \cos\theta & \rho\sin\theta & \rho(1-\cos\theta) \\ -\dfrac{\sin\theta}{\rho} & \cos\theta & \sin\theta \\ 0 & 0 & 1 \end{pmatrix}. \tag{A.3.40}$$

In the vertical plane, neglecting edge focusing, the equation $y'' = 0$ integrates directly to

$$y(s) = y_0 + s\,p_{y0}, \qquad p_y(s) = p_{y0}, \tag{A.3.41}$$

so that the vertical block is simply

$$R^{(\text{dip})}_y = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}. \tag{A.3.42}$$

The longitudinal entries depend on the chosen longitudinal variables and on whether explicit path-length terms are retained.

A.3.7 Drift as a limiting case

A drift of length L is recovered when the focusing functions vanish. In that case,

$$R^{(\text{drift})}_x = R^{(\text{drift})}_y = \begin{pmatrix} 1 & L \\ 0 & 1 \end{pmatrix}, \tag{A.3.43}$$

with the longitudinal block depending on the convention adopted for (ζ, δ).

Annex A: Space Charge and Impedance

Space charge (Vlasov–Poisson). In the beam frame, $\nabla^2\phi = -\rho/\varepsilon_0$ with $\rho = q\int f\,d\mathbf{p}$ and $\mathbf{E}_{sc} = -\nabla\phi$. A symplectic split for each step is

$$\underbrace{e^{:H_{\text{ext}}:\Delta s}}_{\text{drift/external}} \circ \underbrace{e^{:q\phi(\rho):\Delta s}}_{\text{space-charge kick}},$$

where ϕ is computed from f (PIC deposition $\rightarrow$ Poisson solve $\rightarrow$ interpolation). Cost per step adds $O(N_g \log N_g)$ (FFT Poisson) or $\tilde{O}(N_g)$ (multigrid) on N_g grid cells, plus $O(N)$ deposition/interpolation.

In the beam frame, solve $-\nabla^2\phi(\mathbf{x}) = \rho(\mathbf{x})/\varepsilon_0$ on a Cartesian or cylindrical grid, compute $\mathbf{E}_{sc} = -\nabla\phi$, and apply the canonical kick $e^{:q\phi:\Delta s}$. For Gaussian beams, approximate analytic rms forces give envelope equations; for KV, linear self-fields make rms envelopes exact in linear channels. In PIC, the per-step cost is deposition ($O(N)$), Poisson solve ($O(N_g \log N_g)$ FFT or $\tilde{O}(N_g)$ multigrid), and interpolation ($O(N)$).

Annex B: Computational Cost and Error Scaling

Particles (degree n map). Per component, a degree-n polynomial in 6 variables uses $\binom{6+d-1}{d}$ monomials of degree d. Thus per *output*: linear = 6, quadratic = 21, cubic = 56. For six outputs the multiply–adds scale like 36, 126, 336 per particle and step, respectively. Total is $O(kN)$ with degree-dependent constants.

Covariance (Σ). Linear step: $R\Sigma R^\top$ costs $O(6^3)$ per step (constant). Quadratic- or cubic-step additions add fixed tensor contractions (constant). If Σ_0 was estimated from N particles, Twiss uncertainties scale as $O(N^{-1/2})$ and suffer no further Monte-Carlo noise thereafter.

Self-field and wake overhead. Add $O(N_g \log N_g)$ (Poisson) and $O(N_b \log N_b)$ (FFT wakes) per step for N_g grid cells and N_b longitudinal bins.

Wakefields/impedance (nonlocal). Longitudinal force depends on the *history* of the line density,

$$F_\parallel(z,s) = \int_{-\infty}^{s} W_\parallel(z-z', s-s')\,\lambda(z',s')\,ds',$$

equivalently $\widehat{F_\parallel}(\omega) = Z_\parallel(\omega)\,\widehat{\lambda}(\omega)$. This memory is nonlocal in s and does not admit a simple single-particle Hamiltonian unless the state is augmented with auxiliary filter variables. In practice, FFT-based wake kicks are interleaved with symplectic external/self steps; energy/area diagnostics track non-symplectic drift.

Annex C: Coordinates, Laplacians, and Conventions

6D canonical coordinates. We use $\mathbf{z} = (x, p_x, y, p_y, \zeta, \delta)^\top$ with (x, p_x) and (y, p_y) canonical; longitudinally, either (ζ, δ) with ζ a canonical path-like variable and $\delta = \Delta p/p_0$, or $(t, \Delta E)$ with canonical normalization.

Laplacian operators. In Cartesian: $\nabla^2 = \partial_x^2 + \partial_y^2 + \partial_z^2$. In cylindrical (r, θ, z): $\nabla^2 = \partial_r^2 + \frac{1}{r}\partial_r + \frac{1}{r^2}\partial_\theta^2 + \partial_z^2$. In 2D polar (r, θ): $\nabla^2 = \partial_r^2 + \frac{1}{r}\partial_r + \frac{1}{r^2}\partial_\theta^2$.

Annex D (Draft): Thermionic Emission Model

Draft note. This source-model material may later be moved to a dedicated UH accelerator modeling note.

The Richardson–Dushman thermionic emission model including Schottky lowering is written as

$$J = A_R T^2 \exp\left[-\frac{\phi - \Delta\phi_S}{k_B T}\right] \tag{A.7.1}$$

where the Schottky lowering term is

$$\Delta\phi_S = \sqrt{\frac{e^3 E}{4\pi\varepsilon_0}} \tag{A.7.2}$$

Combining both expressions gives

$$J = A_R T^2 \exp\left[-\frac{1}{k_B T}\left(\phi - \sqrt{\frac{e^3 E}{4\pi\varepsilon_0}}\right)\right] \tag{A.7.3}$$

If you want the emitted beam current instead of current density, you can write

$$I = S\, A_R T^2 \exp\left[-\frac{1}{k_B T}\left(\phi - \sqrt{\frac{e^3 E}{4\pi\varepsilon_0}}\right)\right] \tag{A.7.4}$$

where:

$$\begin{aligned}
J &: \text{emitted current density} \\
I &: \text{emitted beam current} \\
A_R &: \text{Richardson constant} \\
T &: \text{cathode temperature} \\
\phi &: \text{work function} \\
\Delta\phi_S &: \text{Schottky barrier lowering} \\
k_B &: \text{Boltzmann constant} \\
e &: \text{elementary charge} \\
E &: \text{surface electric field} \\
\varepsilon_0 &: \text{vacuum permittivity} \\
S &: \text{emitting area}
\end{aligned}$$

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Part B

Beam Transport Optimization

Scope of Part B. This part formulates beam transport as an inverse, exploration, and optimization problem in the space of controllable machine parameters. The basic object is a pullback objective

$$\Phi : \mathcal{K} \subset \mathbb{R}^n \longrightarrow \mathbb{R},$$

where $\mathcal{K}$ is the admissible control hypercube and $\Phi(\mathbf{k})$ is induced by the forward beam-transport map developed in Part A. The central question is not only how to find a small value of Φ, but how to understand the valleys, basins, stiff directions, degeneracies, feasibility boundaries, and phase-advance branches that organize the matching landscape.

The UHM Diagnostic Chicane benchmark motivates this separation. Low-dimensional staged matching can be solved effectively by deterministic local optimizers, while high-dimensional simultaneous matching benefits from Sobol histories and Bayesian/TuRBO-type global search [39]. The strategy developed here therefore separates two layers. The first is a parallel, physics-rich pre-map of the objective topology, including optics labels, response matrices, Hessian spectra, phase advances, dispersion, beta-beating, feasibility, and local-stability information. The second is an adaptive optimizer layer that uses this pre-map to choose initial points, trust-region metrics, active subspaces, surrogate-model histories, or neural-network basin classifiers. Within this second layer we distinguish Bayesian exploration from Bayesian optimization, treat forbidden, probabilistic, and soft constraints separately, and incorporate prior-mean models, kernel correlations, differentiable physics, and trust-region scheduling. This is consistent with accelerator BO demonstrations that combine Gaussian-process surrogates, archived scans, and physics-informed correlations [26, 29]; with machine-learning surrogate strategies for expensive high-fidelity simulations [27]; with differentiable beam-dynamics models such as Cheetah [36]; with prior-mean BO for accelerators [32]; and with recent work on constrained Bayesian exploration and agentic control orchestration [37, 38].

B.1 Inverse Optics and Local Objective Geometry

For a *linear* map, the covariance transports exactly as

$$\Sigma_{\text{out}} = R\,\Sigma_{\text{in}}\,R^{\top}. \tag{B.1.1}$$

For a *quadratic* map

$$z_i' = \sum_j R_{ij} z_j + \frac{1}{2} \sum_{j,k} T_{i\,jk} z_j z_k, \tag{B.1.2}$$

assuming zero mean at the expansion point, one obtains

$$\begin{aligned} \Sigma'_{ab} = (R\Sigma R^{\top})_{ab} &+ \frac{1}{2} \sum_{i,j,m} (R_{ai} T_{b\,jm} + R_{bi} T_{a\,jm})\, \mathbb{E}[z_i z_j z_m] \\ &+ \frac{1}{4} \sum_{i,j,m,n} T_{a\,ij} T_{b\,mn}\, \mathbb{E}[z_i z_j z_m z_n]. \end{aligned} \tag{B.1.3}$$

Gaussian closure (Wick) sets all odd moments to zero and expresses $\mathbb{E}[z_i z_j z_m z_n]$ as sums of products of Σ entries, closing Eq. (B.1.3). For KV/waterbag beams, fourth moments must be computed from uniform-body integrals; Gauss–Hermite closures supply controlled corrections to the Gaussian case.

Cost and accuracy. Let N_p be the number of macroparticles and N_s the number of longitudinal integration steps. Particle tracking costs $O(N_s N_p)$, with constants increasing with map order and collective effects. Twiss estimation from particles has statistical uncertainty of order $O(N_p^{-1/2})$ at each diagnostic location. Covariance transport costs $O(N_s)$ in the linear case, with fixed-size matrix or tensor contractions. If Σ_0 was estimated from N_p particles, its statistical uncertainty is transported deterministically and no additional Monte-Carlo noise is introduced by the transport step itself. See Annex B of Part A.

B.1.1 Inverse use of a fixed linear transport map

Equation (B.1.1) is usually used in the *forward* direction: given the beam matrix at the entrance of a beamline section and the first-order transport matrix R, one computes the beam matrix at the exit. However, when the optics of a section are fixed (for example a chicane with fixed dipole angles and fixed drift lengths), the same relation can be used in the *inverse* direction to determine which upstream beam matrix would be required to obtain prescribed downstream beam parameters.

For a fixed linear map,

$$\mathbf{z}_{\text{out}} = R\,\mathbf{z}_{\text{in}}, \tag{B.1.4}$$

symplecticity implies that R is invertible and $\det R = 1$, so

$$\mathbf{z}_{\text{in}} = R^{-1}\mathbf{z}_{\text{out}}. \tag{B.1.5}$$

Thus, if a desired final coordinate vector $\mathbf{z}^*_{\text{out}}$ is prescribed at the entrance of a downstream device, the corresponding coordinate vector upstream of the fixed section is

$$\mathbf{z}^*_{\text{in}} = R^{-1}\mathbf{z}^*_{\text{out}}. \tag{B.1.6}$$

For second moments, if

$$\Sigma_{\text{out}} = R\Sigma_{\text{in}}R^{\top}, \tag{B.1.7}$$

then

$$\Sigma_{\text{in}} = R^{-1}\Sigma_{\text{out}}R^{-\top}, \tag{B.1.8}$$

and a desired downstream covariance matrix determines the required upstream one:

$$\Sigma^*_{\text{in}} = R^{-1}\Sigma^*_{\text{out}}R^{-\top}. \tag{B.1.9}$$

In a single transverse plane (u, p_u), writing

$$R_u = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \qquad \Sigma_u = \varepsilon_u \begin{pmatrix} \beta_u & -\alpha_u \\ -\alpha_u & \gamma_u \end{pmatrix}, \qquad \gamma_u = \frac{1+\alpha_u^2}{\beta_u}, \tag{B.1.10}$$

the forward Twiss transformation is

$$\beta_{u,\text{out}} = a^2\beta_{u,\text{in}} - 2ab\,\alpha_{u,\text{in}} + b^2\gamma_{u,\text{in}}, \tag{B.1.11}$$

$$\alpha_{u,\text{out}} = -ac\,\beta_{u,\text{in}} + (ad+bc)\alpha_{u,\text{in}} - bd\,\gamma_{u,\text{in}}, \tag{B.1.12}$$

$$\gamma_{u,\text{out}} = c^2\beta_{u,\text{in}} - 2cd\,\alpha_{u,\text{in}} + d^2\gamma_{u,\text{in}}. \tag{B.1.13}$$

Backward propagation is obtained by applying Eqs. (B.1.11)–(B.1.13) to R_u^{-1}.

This inverse use of a fixed map is useful in injection matching: if a beam must enter a downstream section with prescribed Twiss parameters or beam sizes, and the intervening section is fixed, one first computes the required beam matrix at the entrance of the fixed section using Eq. (B.1.9). A tunable upstream matching section can then be optimized to reproduce this upstream target.

It is essential to distinguish this use of R^{-1} from the inverse problem of determining magnet strengths. Equations (B.1.5)–(B.1.9) solve for the *required initial beam state* when the beamline matrix is fixed. They do *not* solve for quadrupole currents, because the currents enter through the dependence

$$R = R(k_1, \dots, k_N), \tag{B.1.14}$$

which makes the matching problem nonlinear in the control variables.

B.1.2 Matching as an inverse problem: response matrices and regularization

In practical optics matching, the unknowns are usually not the initial phase-space coordinates, but the tunable beamline parameters themselves. Let

$$\mathbf{k} = (k_1, \dots, k_n)^\top \in \mathcal{K} \subset \mathbb{R}^n \tag{B.1.15}$$

denote the vector of lattice controls, such as normalized quadrupole strengths or calibrated magnet currents. Let the beamline map depend on these controls,

$$\mathbf{z}_{\text{out}} = \mathcal{M}(\mathbf{z}_{\text{in}}; \mathbf{k}), \tag{B.1.16}$$

with first-order approximation

$$\mathbf{z}_{\text{out}} \approx R(\mathbf{k})\mathbf{z}_{\text{in}}, \qquad \Sigma_{\text{out}} \approx R(\mathbf{k})\Sigma_{\text{in}}R(\mathbf{k})^\top. \tag{B.1.17}$$

If the goal is to reach prescribed observables

$$\mathbf{y}^* \in \mathbb{R}^m, \qquad \mathbf{y}(\mathbf{k}) = (\beta_x, \alpha_x, \beta_y, \alpha_y, D_x, D'_x, \sigma_x, \sigma_y, \dots)^\top, \tag{B.1.18}$$

then the matching problem is

$$\mathbf{y}(\mathbf{k}) = \mathbf{y}^*. \tag{B.1.19}$$

Even in first-order beam transport, this is not linear in $\mathbf{k}$, because quadrupole matrices contain trigonometric or hyperbolic functions of $\sqrt{|k_j|}L_j$ and the full transfer matrix is a product of such element maps.

Local linearization and response matrix. Around a working point $\mathbf{k}_0$, write

$$\mathbf{k} = \mathbf{k}_0 + \Delta\mathbf{k}. \tag{B.1.20}$$

A first-order expansion gives

$$\mathbf{y}(\mathbf{k}_0 + \Delta\mathbf{k}) \approx \mathbf{y}(\mathbf{k}_0) + J_y(\mathbf{k}_0)\Delta\mathbf{k}, \tag{B.1.21}$$

where

$$(J_y)_{ij}(\mathbf{k}_0) = \left.\frac{\partial y_i}{\partial k_j}\right|_{\mathbf{k}_0} \tag{B.1.22}$$

is the optics-response Jacobian. With $\Delta\mathbf{y} = \mathbf{y}^* - \mathbf{y}(\mathbf{k}_0)$, one obtains

$$J_y\Delta\mathbf{k} \approx \Delta\mathbf{y}. \tag{B.1.23}$$

If J_y is square and nonsingular, the correction is $\Delta\mathbf{k} = J_y^{-1}\Delta\mathbf{y}$. More commonly the problem is overdetermined or underdetermined, and one uses the Moore–Penrose pseudoinverse [2, 1]:

$$\Delta\mathbf{k} = J_y^{+}\Delta\mathbf{y}. \tag{B.1.24}$$

This gives the least-squares solution in the overdetermined case and the minimum-norm solution in the underdetermined case.

Singular value decomposition, rank, and numerical conditioning. The pseudoinverse is naturally constructed from the singular value decomposition (SVD) [2, 1]. It is useful to write the full decomposition as

$$J_y = USV^\top = \begin{pmatrix} U_r & U_0 \end{pmatrix} \begin{pmatrix} \Sigma_r & 0 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} V_r & V_0 \end{pmatrix}^\top, \tag{B.1.25}$$

where $J_y \in \mathbb{R}^{m \times n}$, m is the number of selected observables, n is the number of controls, and

$$\Sigma_r = \operatorname{diag}(\sigma_1, \ldots, \sigma_r), \qquad \sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r > 0, \qquad r = \operatorname{rank}(J_y) \leq \min(m, n). \tag{B.1.26}$$

The columns v_j of V_r are orthonormal directions in magnet-current space, and the columns u_j of U_r are orthonormal directions in observable space. Thus a perturbation along one right-singular direction,

$$\Delta\mathbf{k} = a_j v_j, \tag{B.1.27}$$

produces, to first order,

$$\Delta\mathbf{y} \approx J_y \Delta\mathbf{k} = a_j \sigma_j u_j. \tag{B.1.28}$$

The singular value σ_j is therefore the local gain between an orthogonal magnet combination and an orthogonal observable combination. Large σ_j identify strong, well-observed magnet knobs; small σ_j identify weakly observed, poorly conditioned, or nearly degenerate knobs.

The rank r is a central diagnostic of the matching problem. The right null space

$$\mathcal{N}(J_y) = \operatorname{span}\{v_{r+1}, \ldots, v_n\}, \qquad \dim \mathcal{N}(J_y) = n - r, \tag{B.1.29}$$

contains magnet combinations that do not change the selected observables to first order:

$$J_y v_j = 0, \qquad j > r. \tag{B.1.30}$$

These directions correspond to local degeneracies of the matching objective. They may reflect useful lattice freedom, but they may also indicate that the objective is underconstrained. In contrast, the left null space

$$\mathcal{N}(J_y^\top) = \operatorname{span}\{u_{r+1}, \ldots, u_m\}, \qquad \dim \mathcal{N}(J_y^\top) = m - r, \tag{B.1.31}$$

contains observable combinations that cannot be produced, to first order, by any allowed magnet perturbation. Decomposing the desired correction as

$$\Delta\mathbf{y} = U_r U_r^\top \Delta\mathbf{y} + U_0 U_0^\top \Delta\mathbf{y}, \tag{B.1.32}$$

only the component $U_r U_r^\top \Delta\mathbf{y}$ lies in the reachable range of J_y. The unreachable component $U_0 U_0^\top \Delta\mathbf{y}$ remains as a local linear residual.

This gives the following local classification.

- If $m > n$ and $r = n$, the problem is locally overconstrained but all magnet directions are observable; the pseudoinverse gives the least-squares correction.
- If $m < n$ and $r = m$, the selected observables are locally controllable, but the solution is not unique; there are $n - m$ null directions in magnet space.
- If $r < \min(m, n)$, the response is rank deficient: some observable combinations are locally unreachable and some magnet combinations are locally invisible.

For example, if ten quadrupoles are used to match only the four end observables $(\alpha_x, \alpha_y, \beta_x, \beta_y)$, then $J_y \in \mathbb{R}^{4\times 10}$ and $r \leq 4$. Consequently at least six independent current combinations are null to first order, unless additional observables, intermediate constraints, beta-beating penalties, dispersion constraints, aperture constraints, or regularization terms are included.

The Moore–Penrose solution can be written explicitly in modal form as

$$\Delta\mathbf{k}^+ = J_y^+ \Delta\mathbf{y} = \sum_{j=1}^{r} \frac{u_j^\top \Delta\mathbf{y}}{\sigma_j} v_j, \tag{B.1.33}$$

with

$$J_y^+ = V S^+ U^\top, \qquad S^+ = \operatorname{diag}\left(\frac{1}{\sigma_1}, \dots, \frac{1}{\sigma_r}\right). \tag{B.1.34}$$

In an underdetermined problem, the complete local family of solutions is

$$\Delta\mathbf{k} = J_y^+ \Delta\mathbf{y} + \left(I - J_y^+ J_y\right)\mathbf{z}, \qquad \mathbf{z} \in \mathbb{R}^n, \tag{B.1.35}$$

where the second term is an arbitrary displacement in the right null space. This equation makes explicit why a numerical optimizer may find many apparently different quadrupole settings with nearly identical final Twiss parameters: they differ primarily by null-space or near-null-space magnet combinations.

Small singular values amplify measurement noise, model errors, and finite-difference errors. If $\delta\mathbf{y}$ is an error in the target or in the modeled observable, then the component of the correction along v_j scales as

$$\delta a_j \sim \frac{u_j^\top \delta\mathbf{y}}{\sigma_j}. \tag{B.1.36}$$

Thus an ill-conditioned response matrix,

$$\kappa(J_y) = \frac{\sigma_1}{\sigma_r}, \tag{B.1.37}$$

produces narrow valleys and unreliable large corrections. A truncated-SVD or Tikhonov regularization replaces

$$\frac{1}{\sigma_i} \to 0 \qquad \text{for} \qquad \sigma_i < \sigma_{\min}, \tag{B.1.38}$$

thereby discarding directions to which the selected observables are effectively insensitive.

Iterative, damped, and weighted response-matrix updates. The correction in Eq. (B.1.24) is local. After applying it, one recomputes the optics and updates the Jacobian:

$$\mathbf{k}_{\ell+1} = \mathbf{k}_\ell + J_y(\mathbf{k}_\ell)^+ \left(\mathbf{y}^* - \mathbf{y}(\mathbf{k}_\ell)\right). \tag{B.1.39}$$

A damped version uses $0 < \eta \leq 1$,

$$\mathbf{k}_{\ell+1} = \mathbf{k}_\ell + \eta J_y(\mathbf{k}_\ell)^+ \left(\mathbf{y}^* - \mathbf{y}(\mathbf{k}_\ell)\right), \tag{B.1.40}$$

which reduces overshooting in nonlinear regions.

A weighted and regularized local step is obtained by minimizing

$$\mathcal{L}(\Delta\mathbf{k}) = \|W\left(J_y\Delta\mathbf{k} - \Delta\mathbf{y}\right)\|_2^2 + \|\Lambda\Delta\mathbf{k}\|_2^2, \tag{B.1.41}$$

where W weights the observables and Λ penalizes magnet changes. The normal equations are

$$\left(J_y^\top W^\top W J_y + \Lambda^\top\Lambda\right)\Delta\mathbf{k} = J_y^\top W^\top W \Delta\mathbf{y}. \tag{B.1.42}$$

When $\Lambda = \lambda I$, one obtains Tikhonov regularization [2, 1, 3, 4].

B.1.3 Objective design, normalization, and transformed losses

The scalar objective is a modeling choice. It defines which beam properties the optimizer is allowed to trade against each other and which directions in magnet-current space become visible to the Jacobian and Hessian. For a target vector $\mathbf{y}^*$, a general normalized residual may be written

$$\mathbf{r}(\mathbf{k}) = S^{-1}\left(\mathbf{y}(\mathbf{k}) - \mathbf{y}^*\right), \qquad S = \mathrm{diag}(s_1, \dots, s_m), \tag{B.1.43}$$

and the weighted least-squares objective is

$$\Phi(\mathbf{k}) = \frac{1}{\sum_i w_i}\mathbf{r}(\mathbf{k})^\top W \mathbf{r}(\mathbf{k}), \qquad W = \mathrm{diag}(w_1, \dots, w_m). \tag{B.1.44}$$

The scale factors s_i should not be chosen casually. They determine the relative numerical curvature of different observables and therefore influence the conditioning of J_o and H_Φ. Common choices are:

$$\begin{aligned}
s_i &= \Delta y_i^{\mathrm{tol}}, && \text{engineering or physics tolerance,} && \text{(B.1.45)}\\
s_i &= \max\left(|y_i^*|, s_i^{\min}\right), && \text{relative target scaling,} && \text{(B.1.46)}\\
s_i &= \mathrm{std}_{\mathcal{D}}(y_i), && \text{empirical scaling from a Sobol/history set,} && \text{(B.1.47)}\\
s_i &= \sigma_i^{\mathrm{meas}}, && \text{measurement-noise or diagnostic-resolution scaling.} && \text{(B.1.48)}
\end{aligned}$$

If the observables have significant correlations, a more invariant form is the Mahalanobis residual

$$\Phi_C(\mathbf{k}) = \left(\mathbf{y}(\mathbf{k}) - \mathbf{y}^*\right)^\top C_y^{-1}\left(\mathbf{y}(\mathbf{k}) - \mathbf{y}^*\right), \tag{B.1.49}$$

where C_y may represent measurement covariance, model uncertainty, or the empirical covariance of observables in the pre-map. This is equivalent to whitening the output space before computing the response matrix.

In the UHM optimization paper the scalar objective was a weighted MSE with user-defined weights and scales [39]. That form is appropriate for comparing optimizers, but for topology mapping it is useful to store the uncompressed residual vector $\mathbf{r}$ as well as Φ, because two points with the same scalar MSE may fail in physically different ways.

For visualization, basin classification, or BO over objectives spanning many orders of magnitude, one may use a monotone transformed loss

$$\Psi(\mathbf{k}) = \log\left(\Phi(\mathbf{k}) + \epsilon\right), \qquad 0 < \epsilon \ll 1. \tag{B.1.50}$$

The minimizers of Ψ and Φ are identical if ϵ is constant, but the gradients and Hessians are rescaled:

$$\nabla\Psi = \frac{\nabla\Phi}{\Phi + \epsilon}, \tag{B.1.51}$$

$$\nabla^2\Psi = \frac{\nabla^2\Phi}{\Phi + \epsilon} - \frac{\nabla\Phi\nabla\Phi^\top}{(\Phi + \epsilon)^2}. \tag{B.1.52}$$

Thus the logarithmic objective compresses high-error regions and expands the low-error region on plots and in surrogate training. It can help a GP or neural surrogate avoid being dominated by failed optics points, but it also changes local curvature seen by deterministic algorithms. For derivative-based matching near a solution, the untransformed residual form is often more directly connected to the Gauss–Newton geometry; for global pre-mapping and BO history construction, both Φ and $\log(\Phi + \epsilon)$ should be stored.

A complete objective for realistic transport can include physics penalties in addition to the end-point match:

$$\Phi_{\mathrm{total}} = \Phi_{\mathrm{match}} + \rho_\beta P_\beta + \rho_D P_D + \rho_{\mathrm{aper}} P_{\mathrm{aper}} + \rho_{\mathrm{stab}} P_{\mathrm{stab}} + \rho_\kappa P_\kappa, \tag{B.1.53}$$

where the penalties may represent beta beating, dispersion closure, aperture margin, symplectic stability labels, or response-matrix conditioning. These additional terms are also a way to remove null directions when the end-point constraints alone underconstrain the quadrupole vector.

B.1.4 Objective-function curvature: Jacobian, Hessian, and magnet modes

The UHM matching studies use a weighted mean-square objective of the form [39]

$$\Phi(\mathbf{k}) = \frac{1}{\sum_{i=1}^{m} w_i} \sum_{i=1}^{m} w_i \left[\frac{y_i(\mathbf{k}) - y_i^*}{s_i} \right]^2. \tag{B.1.54}$$

It is useful to absorb the scale factors and weights into a residual vector. Define

$$r_i(\mathbf{k}) = \frac{y_i(\mathbf{k}) - y_i^*}{s_i}, \qquad \mathbf{r}(\mathbf{k}) = (r_1, \ldots, r_m)^\top, \qquad W = \mathrm{diag}(w_1, \ldots, w_m). \tag{B.1.55}$$

Ignoring the constant normalization $\sum_i w_i$, the objective is

$$\Phi(\mathbf{k}) = \mathbf{r}(\mathbf{k})^\top W \mathbf{r}(\mathbf{k}). \tag{B.1.56}$$

Let

$$J_o(\mathbf{k}) = \frac{\partial \mathbf{r}}{\partial \mathbf{k}} = \left[\frac{\partial r_i}{\partial k_j} \right] \in \mathbb{R}^{m \times n} \tag{B.1.57}$$

be the Jacobian of the normalized residuals. Then

$$\nabla_{\mathbf{k}} \Phi = 2 J_o^\top W \mathbf{r}, \tag{B.1.58}$$

and the exact Hessian is

$$H_\Phi(\mathbf{k}) = \nabla_{\mathbf{k}}^2 \Phi = 2 J_o^\top W J_o + 2 \sum_{i=1}^{m} w_i r_i(\mathbf{k})\, \nabla_{\mathbf{k}}^2 r_i(\mathbf{k}). \tag{B.1.59}$$

Near a good match, $\mathbf{r} \approx 0$, so the second term is small and the dominant local curvature is the Gauss–Newton Hessian

$$H_{\mathrm{GN}}(\mathbf{k}) = 2 J_o(\mathbf{k})^\top W J_o(\mathbf{k}). \tag{B.1.60}$$

Equivalently, with $\widetilde{J}_o = W^{1/2} J_o$,

$$H_{\mathrm{GN}} = 2 \widetilde{J}_o^\top \widetilde{J}_o. \tag{B.1.61}$$

If

$$\widetilde{J}_o = U \Sigma_J V^\top, \tag{B.1.62}$$

then the Gauss–Newton Hessian has the spectral form

$$H_{\mathrm{GN}} = 2 V \Sigma_J^\top \Sigma_J V^\top, \qquad \lambda_j^{(\mathrm{GN})} = 2\sigma_j^2, \qquad j = 1, \ldots, r. \tag{B.1.63}$$

The remaining $n - r$ Gauss–Newton eigenvalues are exactly zero. Thus the rank of the weighted response matrix is also the rank of the Gauss–Newton curvature model:

$$\mathrm{rank}(H_{\mathrm{GN}}) = \mathrm{rank}(\widetilde{J}_o) = r. \tag{B.1.64}$$

This relation is the central link between response-matrix analysis and objective curvature. A large singular value σ_j produces a large Hessian eigenvalue $2\sigma_j^2$, hence a stiff magnet direction. A small singular value produces a small Hessian eigenvalue, hence a sloppy or weakly observable direction. A null singular value produces a flat direction of the Gauss–Newton model. If

$m < n$, then at least $n - m$ such directions exist unless the objective is enriched with additional observables, intermediate monitors, or physics penalties.

The exact Hessian contains the additional residual-weighted second-derivative term in Eq. (B.1.59). Consequently H_{GN} is always positive semidefinite, whereas H_Φ need not be positive semidefinite away from a solution. The exact Hessian eigenproblem is

$$H_\Phi(\mathbf{k}) v_j(\mathbf{k}) = \lambda_j^{(H)}(\mathbf{k}) v_j(\mathbf{k}). \tag{B.1.65}$$

The signs and magnitudes of $\lambda_j^{(H)}$ classify the local topology of the objective:

$$\lambda_j^{(H)} > 0 \quad \Longrightarrow \quad \text{locally confining curvature along } v_j, \tag{B.1.66}$$

$$\lambda_j^{(H)} < 0 \quad \Longrightarrow \quad \text{downhill direction, ridge, or saddle}, \tag{B.1.67}$$

$$|\lambda_j^{(H)}| \approx 0 \quad \Longrightarrow \quad \text{flat, sloppy, or weakly constrained direction.} \tag{B.1.68}$$

At a strict local minimum all exact Hessian eigenvalues are positive. At a non-isolated minimum some eigenvalues may be zero, indicating a continuous or nearly continuous family of equivalent solutions. At a saddle, at least one eigenvalue is negative. The number of negative eigenvalues,

$$\mathrm{ind}(H_\Phi) = \#\{j : \lambda_j^{(H)} < 0\}, \tag{B.1.69}$$

plays the role of a local Morse index: minima have index zero, first-order saddles have index one, and higher-index saddles lie on more complicated basin boundaries.

The eigenvectors v_j define orthogonal magnet combinations, or modal knobs:

$$\Delta\mathbf{k} = \sum_{j=1}^{n} a_j v_j. \tag{B.1.70}$$

A local quadratic expansion gives

$$\Delta\Phi \approx \frac{1}{2} \Delta\mathbf{k}^\top H_\Phi \Delta\mathbf{k} = \frac{1}{2} \sum_{j=1}^{n} \lambda_j^{(H)} a_j^2. \tag{B.1.71}$$

Therefore the magnitude of the eigenvalue gives the local length scale of the basin along that magnet mode. If one is willing to increase the objective by at most $\Delta\Phi_{\max}$, then a positive-curvature mode should satisfy approximately

$$|a_j| \lesssim \sqrt{\frac{2\Delta\Phi_{\max}}{\lambda_j^{(H)}}}. \tag{B.1.72}$$

Large positive eigenvalues define stiff modes: small current changes rapidly spoil the match. Small positive eigenvalues define sloppy modes: larger current changes are possible with little change in the chosen objective. Nearly zero modes should be interpreted carefully; they may be true operational freedom, or they may reveal that the objective does not constrain enough beam properties. Negative modes should not be treated as basin-floor directions; they indicate where a local descent, restart, or basin-boundary search may be useful.

The Hessian condition number, restricted to the resolved positive spectrum, may be written as

$$\kappa_+(H_\Phi) = \frac{\max\{\lambda_j^{(H)} : \lambda_j^{(H)} > 0\}}{\min\{\lambda_j^{(H)} : \lambda_j^{(H)} > 0\}}, \tag{B.1.73}$$

or regularized by replacing the denominator with $\lambda_j^{(H)} + \epsilon$. Large $\kappa_+(H_\Phi)$ corresponds to a long, narrow valley. This is precisely the regime in which Nelder–Mead can crawl, finite-difference gradient methods can become sensitive to noise, and trust-region algorithms must choose anisotropic step sizes.

A direct optimizer prescription is to rotate from raw currents to Hessian/SVD coordinates:

$$\mathbf{q} = V^\top(\mathbf{k} - \mathbf{k}_0), \qquad \mathbf{k} = \mathbf{k}_0 + V\mathbf{q}. \tag{B.1.74}$$

Here V may be chosen from the SVD of $\widetilde{J}_o$ near a solution, or from the exact Hessian eigensystem when the residual term is important. In these coordinates, the local quadratic model is approximately diagonal. Trust radii may be chosen as

$$|q_j| \le \Delta_j, \qquad \Delta_j \propto \frac{1}{\sqrt{\lambda_j^{(H)} + \epsilon}}, \tag{B.1.75}$$

so that stiff modes receive small allowed steps while sloppy modes may be explored more broadly. For stochastic or population optimizers, the same information can initialize a local covariance

$$C \propto (H_\Phi + \epsilon I)^{-1}, \tag{B.1.76}$$

with regularization $\epsilon > 0$. In a high-dimensional lattice this is also a way to define an active subspace: retain the dominant eigenvectors of H_Φ or $J_o^\top W J_o$ for stiff local matching, and treat the weak modes with broader global exploration, additional physics constraints, or section-wise decomposition.

B.1.5 Computational cost of Jacobian and Hessian information

For n quadrupole variables and m observables, a forward finite-difference Jacobian requires approximately

$$N_{\text{eval}}^{(J,\,\text{forward})} = n + 1 \tag{B.1.77}$$

model evaluations, while a central finite-difference Jacobian requires

$$N_{\text{eval}}^{(J,\,\text{central})} = 2n. \tag{B.1.78}$$

A dense finite-difference Hessian of the scalar objective costs $O(n^2)$ evaluations. For a first-order matrix model this is often acceptable, but for RF-Track, Xsuite, COSY INFINITY, or particle-based tracking the derivative budget must be used selectively.

The Gauss–Newton curvature in Eq. (B.1.60) requires only the Jacobian plus dense linear algebra:

$$\text{cost}(H_{\text{GN}}) \sim O(n) \text{ model evaluations} + O(n^2 m) \text{ linear algebra}. \tag{B.1.79}$$

This is the appropriate first curvature estimate for a large pre-map. A practical hierarchy is: compute cheap optics labels at all sampled points; compute J_o and H_{GN} at promising, representative, or suspicious points; compute the exact Hessian only at candidate minima, saddles, or benchmark points where the second-order residual term in Eq. (B.1.59) may matter.

B.1.6 Phase advance, symplectic eigenvalues, and normal-form coordinates

The first-order map through a Hamiltonian section is symplectic:

$$R(\mathbf{k})^\top \mathbf{J} R(\mathbf{k}) = \mathbf{J}. \tag{B.1.80}$$

Its eigenvalues occur in reciprocal complex-conjugate pairs. For a stable uncoupled plane,

$$\lambda_{u,\pm} = e^{\pm i\mu_u}, \qquad u \in \{x, y\}, \tag{B.1.81}$$

where μ_u is the phase advance through the section. If

$$R_u = \begin{pmatrix} R_{11} & R_{12} \\ R_{21} & R_{22} \end{pmatrix}, \tag{B.1.82}$$

then

$$\cos\mu_u = \frac{1}{2}\,\mathrm{Tr}(R_u). \tag{B.1.83}$$

Equivalently, the Twiss parametrization of a stable 2×2 block is

$$R_u = \begin{pmatrix} \cos\mu_u + \alpha_u \sin\mu_u & \beta_u \sin\mu_u \\ -\gamma_u \sin\mu_u & \cos\mu_u - \alpha_u \sin\mu_u \end{pmatrix}, \qquad \gamma_u = \frac{1+\alpha_u^2}{\beta_u}. \tag{B.1.84}$$

In a coupled 4D or 6D map, one may instead use the symplectic normal-form construction introduced in Part A,

$$R = A\mathcal{R}A^{-1}, \qquad \mathcal{R} = \mathrm{diag}\left(R(\mu_1), R(\mu_2), R(\mu_3)\right), \tag{B.1.85}$$

with

$$R(\mu_j) = \begin{pmatrix} \cos\mu_j & \sin\mu_j \\ -\sin\mu_j & \cos\mu_j \end{pmatrix}. \tag{B.1.86}$$

The columns of A define the coupled normal-mode coordinates and lattice functions. In these coordinates, stable transport is a rotation in each mode rather than a mixed transformation in raw physical coordinates.

This matters for optimization because two different current vectors may satisfy similar final Twiss and dispersion targets while accumulating different internal phase advances:

$$\Phi(\mathbf{k}_a) \approx \Phi(\mathbf{k}_b), \qquad \boldsymbol{\mu}(\mathbf{k}_a) \not\equiv \boldsymbol{\mu}(\mathbf{k}_b) \pmod{2\pi}. \tag{B.1.87}$$

Such points should be treated as distinct optics branches or candidate islands. Phase advance, normal-mode mixing, beta beating, dispersion response, and Hessian conditioning should therefore be stored as labels of a point in current space, not only the scalar objective value. A typical feature vector is

$$\mathcal{F}(\mathbf{k}) = \left(\Phi, \boldsymbol{\mu}, \frac{\Delta\beta_x}{\beta_x}, \frac{\Delta\beta_y}{\beta_y}, D_x, D_x', \mathrm{spec}(H_\Phi), \mathrm{spec}(J_o^\top J_o), \kappa(J_o)\right). \tag{B.1.88}$$

Here spec denotes the eigenvalue or singular-value spectrum. Near $|\,\mathrm{Tr}(R_u)| = 2$, the phase advance approaches 0 or π and the optical response often becomes ill-conditioned; such regions are natural candidates for penalties or constraints. For example,

$$P_{\mathrm{stab}} = \sum_u \left[\max\left(0, |\,\mathrm{Tr}(R_u)| - 2 + \epsilon\right)\right]^2, \qquad P_\kappa = \log\left(1 + \kappa(J_o)^2\right). \tag{B.1.89}$$

The interpretation of the phase and eigenvalue labels depends on the accelerator class. In a circular accelerator, the relevant map is often a one-turn or multi-turn map; the tunes (ν_x, ν_y, ν_s) and resonance conditions

$$\ell_x\nu_x + \ell_y\nu_y + \ell_s\nu_s = N, \qquad \ell_x, \ell_y, \ell_s, N \in \mathbb{Z}, \tag{B.1.90}$$

organize nonlinear islands, tune diffusion, and dynamic aperture. In that regime, normal-form terms and frequency-map analysis provide direct resonance labels for the optimization classifiers. In a single-pass linear accelerator or transport line, such as the UH Diagnostic Chicane, there is no repeated turn-by-turn tune in the same sense. Nevertheless, the finite-section phase advances (μ_x, μ_y) still label different focusing branches because the transfer matrix contains the same sine, cosine, hyperbolic, and dispersion response structure. Multiple current vectors can therefore satisfy the same final Twiss constraints while belonging to different accumulated-phase branches.

Higher-fidelity models can modify these labels. Space charge, wakefields, coherent synchrotron radiation, incoherent synchrotron-radiation energy loss, quantum excitation, magnet fringe fields, and aperture clipping change the forward map and hence change J_o, H_Φ, and their eigenvectors. If radiation or aperture loss is modeled as a dissipative process, the map is no longer exactly symplectic and stability regions predicted by a first-order Hamiltonian model may be damped, shifted, or blurred. In the UH linear DBA chicane this distinction is important when moving from the first-order proof-of-concept model to RF-Track, COSY INFINITY, Xsuite, or particle-based tracking with bending-radiation and collective effects.

B.1.7 High-order maps and consistency of the response

The distinction between first-order and higher-order transport remains essential. Even when transport is first order in phase-space coordinates, the observable vector $\mathbf{y}(\mathbf{k})$ is nonlinear in the controls. At higher order,

$$\mathbf{z}_{\text{out}} = \mathbf{a} + R(\mathbf{k})\mathbf{z}_{\text{in}} + \frac{1}{2}T(\mathbf{k})[\mathbf{z}_{\text{in}}, \mathbf{z}_{\text{in}}] + \frac{1}{3!}U(\mathbf{k})[\mathbf{z}_{\text{in}}, \mathbf{z}_{\text{in}}, \mathbf{z}_{\text{in}}] + \cdots, \tag{B.1.91}$$

where T and U are the Taylor tensors introduced in Eq. (A.2.24). The residual Jacobian remains

$$(J_o)_{ij}(\mathbf{k}) = \frac{\partial r_i}{\partial k_j}, \tag{B.1.92}$$

but the function from which it is computed now includes the sensitivity of the selected observables to $R(\mathbf{k})$, $T(\mathbf{k})$, $U(\mathbf{k})$, the input distribution, and the moment closure used for covariance propagation. Therefore, response matrices, Hessians, phase advances, and normal-form labels must be computed from the same level of physics retained in the forward model: first-order linear matrix, differential-algebraic Taylor map, or particle tracking. Differential algebra, finite differences, algorithmic differentiation, and surrogate models all provide possible routes to these derivatives; for automatic-differentiation-based surrogate and BO implementations, see [30].

B.1.8 Response-based splitting of the beamline into sections

The staged optimization used in the Diagnostic Chicane benchmark [39] can be understood through the block structure of the response matrix. Partition the quadrupole vector and observables into sections,

$$\mathbf{k} = \left(\mathbf{k}^{(1)}, \ldots, \mathbf{k}^{(S)}\right), \qquad \mathbf{y} = \left(\mathbf{y}^{(1)}, \ldots, \mathbf{y}^{(S)}\right). \tag{B.1.93}$$

Then

$$J_o = \begin{pmatrix} J_{11} & J_{12} & \cdots & J_{1S} \\ J_{21} & J_{22} & \cdots & J_{2S} \\ \vdots & \vdots & \ddots & \vdots \\ J_{S1} & J_{S2} & \cdots & J_{SS} \end{pmatrix}. \tag{B.1.94}$$

A section-wise split is physically justified when the diagonal response blocks dominate:

$$\|J_{ss}\| \gg \|J_{sr}\|, \qquad r \neq s. \tag{B.1.95}$$

Equivalently, the Gauss–Newton Hessian has approximate block structure,

$$H_{\text{GN}} \approx \begin{pmatrix} H_{11} & \epsilon_{12} & \cdots \\ \epsilon_{21} & H_{22} & \cdots \\ \vdots & \vdots & \ddots \end{pmatrix}, \qquad \|\epsilon_{sr}\| \ll \|H_{ss}\|. \tag{B.1.96}$$

If off-diagonal blocks are large, the sections are strongly coupled and should be merged or optimized with overlap. If a section has very small singular values, it is underconstrained and needs additional observables, intermediate targets, beta-beating penalties, dispersion closure, or aperture constraints.

The support of a Hessian eigenvector across sections is also diagnostic. For eigenvector v_j, define

$$\chi_{j,s} = \sum_{\ell \in s} |(v_j)_\ell|^2, \qquad \sum_{s=1}^{S} \chi_{j,s} = 1. \tag{B.1.97}$$

If $\chi_{j,s}$ is localized in one section, the corresponding magnet combination can be optimized locally. If $\chi_{j,s}$ is distributed over several adjacent sections, the corresponding knob is a coupled mode and the split should be modified. This provides a quantitative criterion for choosing staged, overlapping, or full-space matching.

B.2 Topology Mapping of the Matching Landscape

The previous section describes local inverse matching and the local differential geometry of the objective. The present section describes the global strategy suggested by that structure. The central idea is to separate the problem into two layers:

1. a *pre-mapping layer*, in which the admissible hypercube $\mathcal{K} \subset \mathbb{R}^n$ is sampled in parallel and each point is labeled by objective, optics, response, curvature, and spectral information;
2. an *optimizer layer*, in which deterministic, population-based, or Bayesian algorithms use the pre-map as initial conditions, constraints, local coordinates, covariance matrices, or history data.

Sobol sampling is therefore not the complete strategy; it is the null hypothesis for the pre-map. The physics information attached to each sampled point is what turns a Sobol history into a topology map of the matching landscape.

B.2.1 Basins, islands, and the pullback objective

The scalar matching landscape is the pullback of the beam-transport map onto magnet-current space:

$$\Phi : \mathcal{K} \subset \mathbb{R}^n \to \mathbb{R}, \qquad \Phi(\mathbf{k}) = \Phi[\mathcal{M}(\cdot;\mathbf{k})]\,. \tag{B.2.1}$$

The islands observed in a scan of $\mathcal{K}$ are not, in general, Hamiltonian phase-space islands. They are basins of attraction of the optimizer flow induced by the pullback objective. For the gradient flow

$$\dot{\mathbf{k}} = -\nabla\Phi(\mathbf{k}), \tag{B.2.2}$$

the basin of a local minimum $\mathbf{k}_a^*$ is

$$\mathcal{B}_a = \left\{\mathbf{k}_0 \in \mathcal{K} : \lim_{t\to\infty} \varphi_t(\mathbf{k}_0) = \mathbf{k}_a^*\right\}, \tag{B.2.3}$$

where φ_t is the flow map. Basin boundaries are organized by saddle manifolds of Φ. The Morse index

$$\mathrm{ind}(\mathbf{k}) = \#\{j : \lambda_j^{(H)}(\mathbf{k}) < 0\} \tag{B.2.4}$$

classifies the local critical structure: minima have index zero, saddles have positive index, and nearly zero eigenvalues indicate flat or degenerate directions. This is the continuum analogue of a Morse–Smale decomposition, which in data analysis partitions scalar fields into regions of equivalent gradient-flow behavior and simplifies them by persistence of critical points [20].

B.2.2 Parallel Sobol pre-mapping with optics and curvature labels

Let

$$\mathcal{S}_{N_q} = \{\mathbf{k}^{(p)}\}_{p=1}^{N_q} \subset \mathcal{K} \tag{B.2.5}$$

be a Sobol or other low-discrepancy sample of the admissible current hypercube [15]. In a first-order matrix model, the evaluations are cheap and embarrassingly parallel; in practice one may target $N_q \sim 10^5$–10^6 points, then restrict derivative or high-order diagnostics to a smaller subset. At each sampled point, the proposed database is

$$\mathcal{D}_{\mathrm{map}} = \left\{\mathbf{k}^{(p)}, \Phi^{(p)}, \mathbf{r}^{(p)}, \boldsymbol{\mu}^{(p)}, \Delta\beta^{(p)}/\beta, D_x^{(p)}, D_x'^{(p)}, J_o^{(p)}, H_{\mathrm{GN}}^{(p)}, \mathrm{spec}(H_{\mathrm{GN}}^{(p)}), \kappa(J_o^{(p)})\right\}_{p=1}^{N_q}. \tag{B.2.6}$$

For high-fidelity tracking, one can use a tiered strategy:

$$\mathcal{D}_0: \quad \{\Phi, \mathbf{r}, \boldsymbol{\mu}, \Delta\beta/\beta, D_x, D'_x\}, \qquad \text{all } N_q \text{ points}, \tag{B.2.7}$$

$$\mathcal{D}_1: \quad \{J_o, H_{\mathrm{GN}}, \mathrm{spec}(H_{\mathrm{GN}}), \kappa(J_o)\}, \qquad \text{selected } N_1 \ll N_q, \tag{B.2.8}$$

$$\mathcal{D}_2: \quad \{H_\Phi \text{ exact, nonlinear maps, robustness tests}\}, \qquad \text{selected } N_2 \ll N_1. \tag{B.2.9}$$

The map is then no longer a list of objective values. It is a structured data set describing which optics branches, response ranks, Hessian signatures, and beta/dispersion patterns are present in the hypercube.

B.2.3 Graph descent and basin classification in augmented feature space

A practical basin classifier can be built without gridding the full n-dimensional hypercube. Construct a k-nearest-neighbor graph on the sampled points using a feature-weighted distance, for example

$$\begin{aligned} d^2(p,q) = {}& \left\|\mathbf{k}^{(p)} - \mathbf{k}^{(q)}\right\|_A^2 + a_\mu \left\|\mathrm{wrap}_{2\pi}\left(\boldsymbol{\mu}^{(p)} - \boldsymbol{\mu}^{(q)}\right)\right\|_2^2 \\ & + a_H \left\|\log\left(|\boldsymbol{\lambda}_H^{(p)}| + \epsilon\right) - \log\left(|\boldsymbol{\lambda}_H^{(q)}| + \epsilon\right)\right\|_2^2 + a_D \left\|\mathbf{D}^{(p)} - \mathbf{D}^{(q)}\right\|_2^2. \end{aligned} \tag{B.2.10}$$

Each point is then assigned to the neighbor with lower objective,

$$p \longmapsto \arg\min_{q \in \mathcal{N}(p)} \Phi^{(q)}, \tag{B.2.11}$$

and the process is iterated until a discrete sink is reached. Points that flow to the same sink form a candidate basin. Candidate basins are merged only if their local-refined minima, phase advances, Hessian signatures, and physical optics labels agree within tolerances. This procedure is a discrete analogue of following the gradient flow in Eq. (B.2.2).

A basin label may be written

$$b(\mathbf{k}) \in \{1, \ldots, N_{\mathrm{basin}}\}. \tag{B.2.12}$$

The classification should not be based only on the value of Φ. Two points may have similar objective values but lie on different phase-advance branches or have different internal beta beating. Conversely, points separated in raw current space may belong to the same basin if their feature vectors and local-refined minima agree.

B.2.4 Targeted Fourier and frequency analysis along physics directions

There are two distinct spectral analyses. The first is the eigenvalue spectrum of R, J_o, or H_Φ, already discussed. The second is Fourier or frequency analysis of a scalar signal sampled along a direction in magnet space. A full n-dimensional FFT of $\Phi(k_1, \ldots, k_n)$ is not practical for

$n \sim$ 10–16, because a grid with M points per variable would require M^n evaluations. The useful object is instead a one- or two-dimensional slice along physics-informed directions.

Let d_j be a direction in current space. Natural choices are:

$$d_j \in \left\{ \begin{array}{l} \text{individual quadrupole axes,} \\ \text{right singular vectors of } J_o, \\ \text{eigenvectors of } H_\Phi, \\ \text{section knobs or phase-advance knobs} \end{array} \right\}. \tag{B.2.13}$$

Define a line scan

$$\mathbf{k}(s) = \mathbf{k}_0 + s d_j, \qquad s \in [s_{\min}, s_{\max}], \tag{B.2.14}$$

and compute

$$\Phi_j(s) = \Phi(\mathbf{k}(s)), \qquad \mu_{u,j}(s) = \mu_u(\mathbf{k}(s)), \qquad \lambda^{(H)}_{\ell,j}(s) = \lambda^{(H)}_\ell(\mathbf{k}(s)). \tag{B.2.15}$$

A windowed discrete Fourier transform of $\Phi_j(s)$,

$$\widehat{\Phi}_j(\omega) = \sum_{r=0}^{M-1} w_r \Phi_j(s_r) e^{-i\omega s_r}, \tag{B.2.16}$$

reveals dominant repeat lengths along that magnet mode. If a dominant frequency ω_j^* exists, the characteristic repeat distance is

$$\ell_j \approx \frac{2\pi}{\omega_j^*}, \tag{B.2.17}$$

and the expected number of oscillations over a scan length $L_j = s_{\max} - s_{\min}$ is roughly

$$N^{(j)}_{\text{osc}} \sim \frac{\omega_j^* L_j}{2\pi}. \tag{B.2.18}$$

This does not count all basins in $\mathbb{R}^n$, but it estimates repeatability and basin spacing along physically meaningful directions. If the sampling is nonuniform, one may use nonuniform spectral estimation rather than a standard FFT.

This approach is related in spirit to frequency map analysis, which uses refined Fourier analysis to study Hamiltonian or symplectic maps and is widely used in accelerator dynamics [17, 18, 19]. In the present single-pass matching problem, the goal is different: we do not analyze turn-by-turn particle tunes. Instead, we analyze how Φ, $\boldsymbol{\mu}$, and the Hessian/Jacobian spectra vary as functions of magnet modes. For example, if $\lambda_{\max}(H_\Phi(\mathbf{k}(s)))$ peaks whenever

$$\mu_x(s) \approx m\pi \quad \text{or} \quad \mu_y(s) \approx n\pi, \tag{B.2.19}$$

then the optimizer should expect stiff, narrow valleys or basin boundaries near those phase-advance branches. Conversely, a repeated low-Φ pattern as a function of (μ_x, μ_y) indicates that several local minima may be copies of the same matching condition on different accumulated-phase branches.

A more physical spectral map is therefore not

$$\Phi(k_1, \ldots, k_n), \tag{B.2.20}$$

but rather

$$\Phi\left(\mu_x, \mu_y, \Delta\beta/\beta, D_x, D'_x\right), \tag{B.2.21}$$

or its projection onto the active Hessian/SVD coordinates. Active-subspace methods formalize this idea by using eigenvectors of averaged gradient outer products to identify important parameter-space directions for high-dimensional studies [16].

B.2.5 Neural-network layer on top of the physics classifiers

The pre-map can also train a neural-network or other machine-learning classifier. The input is not only the current vector, but an augmented physics feature vector:

$$\mathbf{x}(\mathbf{k}) = \Big(\mathbf{k}, \Phi, \boldsymbol{\mu}, \Delta\beta/\beta, D_x, D'_x, \mathrm{spec}(H_{\mathrm{GN}}), \mathrm{spec}(J_o^\top J_o), \kappa(J_o), \mathrm{ind}(H_\Phi)\Big). \tag{B.2.22}$$

The targets may be basin labels from graph descent or local restarts,

$$C_\theta(\mathbf{x}) \approx P(b = a \mid \mathbf{x}), \tag{B.2.23}$$

objective or observable surrogates,

$$R_\theta(\mathbf{x}) \approx \Phi(\mathbf{k}) \quad \text{or} \quad \widehat{\mathbf{y}}_\theta(\mathbf{k}) \approx \mathbf{y}(\mathbf{k}), \tag{B.2.24}$$

or geometry classifiers such as $\lambda_{\min}(H)$, $\kappa(J_o)$, or the Morse index. The neural layer should not be asked to rediscover accelerator physics from raw currents alone; it should be conditioned on phase advance, beta beating, dispersion, and response/Hessian features.

Once trained, the classifier can be used for active learning. New candidate points are selected because they are predicted to lie in a good basin, because the classifier is uncertain about their basin label, or because they lie near a predicted basin boundary. A schematic acquisition score is

$$S(\mathbf{k}) = -\widehat{\Phi}(\mathbf{k}) + a\, U_b(\mathbf{k}) + b\, P(\text{good basin} \mid \mathbf{k}) - c\, P_{\mathrm{stab}}(\mathbf{k}) - d\, P_\kappa(\mathbf{k}), \tag{B.2.25}$$

where U_b is classifier uncertainty. This produces a feedback loop:

$$\text{Sobol/pre-map} \to \text{labels} \to \text{classifier} \to \text{targeted new samples} \to \text{updated topology map}. \tag{B.2.26}$$

B.3 Optimization and the UH Workflow

The topology map of Section B.2 is not itself an optimizer. It is a structured prior for choosing where and how to optimize. This section separates the algorithmic layer from the mapping layer. Deterministic algorithms, population methods, Bayesian exploration, BO, TuRBO, and neural surrogates can all be used; the main UH strategy is to make them less blind by supplying basin labels, modal coordinates, trust-region metrics, feasibility information, solution-stability measures, and physics-informed histories.

B.3.1 Deterministic optimization algorithms

Consider the bound-constrained scalar problem

$$\min_{\mathbf{k} \in \mathcal{K} \subset \mathbb{R}^n} \Phi(\mathbf{k}), \qquad \mathcal{K} = \prod_{j=1}^{n} [k_j^{\min}, k_j^{\max}], \tag{B.3.1}$$

possibly supplemented by equality and inequality constraints

$$h_a(\mathbf{k}) = 0, \qquad g_b(\mathbf{k}) \le 0. \tag{B.3.2}$$

The SciPy interface `optimize.minimize` provides many algorithms for such problems [12, 13, 14]. They differ mainly in whether they require derivatives, whether they handle bounds or nonlinear constraints, and whether they use line searches, trust regions, simplex geometry, or sequential quadratic approximations.

B.3.1.1 Derivative-free local methods: Nelder–Mead and Powell

Nelder–Mead maintains a simplex with $n+1$ vertices

$$\mathcal{V}_\ell = \{\mathbf{k}_1, \ldots, \mathbf{k}_{n+1}\} \tag{B.3.3}$$

and updates the worst vertex using reflection, expansion, contraction, and shrink steps. If the vertices are ordered so that

$$\Phi(\mathbf{k}_1) \leq \cdots \leq \Phi(\mathbf{k}_{n+1}), \tag{B.3.4}$$

and $\bar{\mathbf{k}}$ is the centroid of all but the worst vertex, the reflection point is

$$\mathbf{k}_r = \bar{\mathbf{k}} + \alpha(\bar{\mathbf{k}} - \mathbf{k}_{n+1}). \tag{B.3.5}$$

Expansion and contraction use analogous affine combinations. Nelder–Mead is useful when derivatives are unavailable and the dimension is small, but it does not exploit the response matrix and can stagnate in narrow valleys. Powell's method is also derivative-free; it performs sequential line minimizations along a set of directions that are updated during the search. In beam matching, these methods are most appropriate as local refiners for a few knobs or as baseline black-box methods.

B.3.1.2 Finite-difference and quasi-Newton methods: BFGS and L-BFGS-B

When a gradient is available or can be estimated, quasi-Newton methods approximate local curvature without forming the exact Hessian. BFGS updates an approximate Hessian B_k using

$$B_{k+1} = B_k + \frac{\mathbf{y}_k \mathbf{y}_k^\top}{\mathbf{y}_k^\top \mathbf{s}_k} - \frac{B_k \mathbf{s}_k \mathbf{s}_k^\top B_k}{\mathbf{s}_k^\top B_k \mathbf{s}_k}, \tag{B.3.6}$$

where

$$\mathbf{s}_k = \mathbf{k}_{k+1} - \mathbf{k}_k, \qquad \mathbf{y}_k = \nabla\Phi(\mathbf{k}_{k+1}) - \nabla\Phi(\mathbf{k}_k). \tag{B.3.7}$$

L-BFGS-B stores only a limited number of $(\mathbf{s}_k, \mathbf{y}_k)$ pairs and handles simple bounds. It is attractive for large numbers of variables when gradients are cheap, but it remains local and can follow the wrong valley if initialized outside a good basin [5, 10].

B.3.1.3 Newton and trust-region methods

Newton's method uses the local quadratic model

$$m_k(\mathbf{p}) = \Phi(\mathbf{k}_k) + \nabla\Phi(\mathbf{k}_k)^\top \mathbf{p} + \frac{1}{2}\mathbf{p}^\top H_k \mathbf{p} \tag{B.3.8}$$

and proposes $\mathbf{p}_k = -H_k^{-1}\nabla\Phi(\mathbf{k}_k)$ when H_k is positive definite. Trust-region methods instead solve

$$\min_{\|\mathbf{p}\| \leq \Delta_k} m_k(\mathbf{p}), \tag{B.3.9}$$

where the radius Δ_k is expanded or contracted according to the agreement between the model and the true objective. Algorithms such as `trust-ncg`, `trust-krylov`, `dogleg`, `trust-exact`, and `trust-constr` differ in how they solve Eq. (B.3.9) and how they handle constraints [6, 5, 13]. In the matching problem, the Hessian/SVD basis from Section B.1.4 suggests an anisotropic trust region,

$$\mathbf{p}^\top (H_\Phi + \epsilon I)\mathbf{p} \leq \Delta_k^2, \tag{B.3.10}$$

which is more natural than an isotropic Euclidean ball when the optics response is ill-conditioned.

B.3.1.4 Constrained local methods: SLSQP, COBYLA, and COBYQA

Sequential least-squares programming (SLSQP) is a sequential quadratic programming method. At iteration k it solves a quadratic subproblem of the form

$$\begin{aligned} \min_{\mathbf{p}} \quad & \nabla\Phi(\mathbf{k}_k)^\top \mathbf{p} + \frac{1}{2}\mathbf{p}^\top B_k \mathbf{p} \\ \text{s.t.} \quad & h(\mathbf{k}_k) + Dh(\mathbf{k}_k)\mathbf{p} = 0, \\ & g(\mathbf{k}_k) + Dg(\mathbf{k}_k)\mathbf{p} \leq 0, \end{aligned} \tag{B.3.11}$$

where B_k approximates the Hessian of the Lagrangian. SLSQP is effective when derivative information is reasonably accurate and constraints such as magnet limits, dispersion bounds, or aperture margins are important. COBYLA and COBYQA are derivative-free constrained methods based on local linear or quadratic approximations. They are useful when the objective is black-box or noisy, but they do not directly exploit the symplectic response structure.

B.3.1.5 Role in the UHM benchmark

The UHM Diagnostic Chicane benchmark compared Nelder–Mead, COBYLA, L-BFGS-B, SLSQP, and `trust-constr` on three matching scenarios [39]. Scenario A was a two-quadrupole map with several local minima; Scenario B decomposed the Diagnostic Chicane into twelve staged low-dimensional matches; Scenario C optimized a coupled high-dimensional quadrupole set simultaneously. The conclusion was that local methods are reliable for small or staged problems, SLSQP with finite-difference Jacobians is efficient in the staged case, and purely local methods become unreliable in the full high-dimensional case. This motivates the topology pre-map and BO/TuRBO layer.

Method class	Derivatives	Constraints	Comments for beam matching
Nelder–Mead	none	weak / bounds by wrappers	Simple baseline; useful for few knobs; poor in narrow valleys.
Powell	none	bounds in implementations	Directional line searches; may work for smooth low-dimensional scans.
L-BFGS-B	gradient	box bounds	Efficient for many variables if gradients are reliable; local.
Newton-CG / TNC	gradient, Hessian-vector products	bounds for TNC	Good when Hessian-vector products are available from AD or model derivatives.
Trust-region methods	gradient and Hessian or Hessian approximation	varies	Natural when Hessian/SVD basis supplies a metric and trust radii.
SLSQP	gradient/Jacobian	bounds and nonlinear constraints	Good for staged constrained optics matching; used in the UHM benchmark.
COBYLA/COBYQA	none	nonlinear constraints	Useful for black-box constrained tuning; may require many evaluations.

Table B.1: Deterministic optimizer families relevant to beam matching. The distinction between derivative-free and derivative-informed methods is central when moving from a first-order map, where derivatives are cheap, to a high-fidelity or online model, where evaluations may be expensive or noisy.

B.3.2 Bayesian optimization and exploration

Bayesian optimization (BO) targets expensive, noisy, derivative-free optimization problems. Let

$$z_i = f(\mathbf{k}_i) + \epsilon_i, \qquad \epsilon_i \sim \mathcal{N}(0, \sigma_n^2), \tag{B.3.12}$$

where f may be Φ, $\log(\Phi + \epsilon)$, or a scalar machine-performance metric. In Gaussian-process (GP) BO one assumes

$$f(\mathbf{k}) \sim \mathcal{GP}\left(m(\mathbf{k}), K(\mathbf{k}, \mathbf{k}')\right), \tag{B.3.13}$$

where the mean m carries prior knowledge and the kernel K defines covariance, smoothness, characteristic length scales, and correlations. Given observations $\mathbf{z}$ at points X, the posterior at a new point $\mathbf{k}$ is

$$\mu(\mathbf{k}) = m(\mathbf{k}) + K_{\mathbf{k}X}\left(K_{XX} + \sigma_n^2 I\right)^{-1}(\mathbf{z} - m_X), \tag{B.3.14}$$

$$\sigma^2(\mathbf{k}) = K(\mathbf{k},\mathbf{k}) - K_{\mathbf{k}X}\left(K_{XX} + \sigma_n^2 I\right)^{-1} K_{X\mathbf{k}}. \tag{B.3.15}$$

The acquisition function converts this posterior into the next point to evaluate,

$$\mathbf{k}_{n+1} = \arg\max_{\mathbf{k}\in\mathcal{K}} a_n(\mathbf{k}). \tag{B.3.16}$$

For maximization, the upper confidence bound is

$$a_{\mathrm{UCB}}(\mathbf{k}) = \mu(\mathbf{k}) + \kappa\sigma(\mathbf{k}), \tag{B.3.17}$$

while for minimization one uses the corresponding lower confidence bound or applies UCB to $-\Phi$. Expected improvement for minimization can be written

$$a_{\mathrm{EI}}(\mathbf{k}) = (f_{\mathrm{best}} - \mu)\Phi_N\left(\frac{f_{\mathrm{best}} - \mu}{\sigma}\right) + \sigma\phi_N\left(\frac{f_{\mathrm{best}} - \mu}{\sigma}\right), \tag{B.3.18}$$

where Φ_N and ϕ_N are the standard normal CDF and PDF. UCB emphasizes optimism under uncertainty; EI emphasizes expected gain relative to the best observed value [21, 22, 23, 24].

B.3.2.1 Exploration and exploitation

Bayesian *exploration* uses the same probabilistic model for a different purpose. Rather than immediately minimizing Φ, it seeks to improve knowledge of the response map, the feasible set, or the basin structure. In the UH framework this may mean: (i) exploring poorly sampled regions with large posterior variance; (ii) refining a boundary between feasible and infeasible transport; (iii) populating under-sampled candidate basins discovered by the Sobol pre-map; or (iv) learning an observable surrogate before solving the final matching problem. Bayesian *optimization* is the exploitative stage that searches for high-performing settings using the model built during exploration.

The balance can be adjusted continuously. EI and small-κ UCB favor exploitation; variance-based criteria and larger κ favor exploration. A physics-informed composite score may combine predicted objective, uncertainty, basin probability, and penalties,

$$S(\mathbf{k}) = -\widehat{\Phi}(\mathbf{k}) + a\,U_b(\mathbf{k}) + b\,P(\text{good basin} \mid \mathbf{k}) - c\,P_{\mathrm{stab}}(\mathbf{k}) - d\,P_{\kappa}(\mathbf{k}), \tag{B.3.19}$$

where U_b is classifier uncertainty, P_{stab} penalizes predicted instability, and P_κ penalizes ill-conditioned response. This makes the transition from topology discovery to targeted exploitation explicit rather than implicit.

B.3.2.2 Surrogate models

The surrogate should be distinguished from the optimizer. A surrogate approximates the forward map or objective; an optimizer uses that model to choose settings. The local linear surrogate is

$$\widehat{\mathbf{y}}(\mathbf{k}) = \mathbf{y}(\mathbf{k}_0) + J_y(\mathbf{k}_0)(\mathbf{k} - \mathbf{k}_0). \tag{B.3.20}$$

Polynomial regression, polynomial-chaos expansions, support-vector regression, GPs, neural networks, and differentiable physics models extend this idea to larger regions of $\mathcal{K}$. GPs provide calibrated posterior uncertainty for BO, but exact models scale poorly with very large histories unless sparse or approximate methods are used. Neural networks are fast at inference and scale well to large data sets, but uncertainty must be supplied through ensembles, Bayesian neural networks, dropout, or calibration. When a physics or learned surrogate is differentiable, automatic differentiation can provide gradients, Jacobians, Hessian-vector products, and derivative observations that connect directly to the response and curvature theory of Section B.1.4. These derivatives may inform local refinement, SVD coordinates, kernel metrics, and trust-region geometry without replacing the probabilistic uncertainty model. High-fidelity accelerator surrogates can reduce evaluation cost by many orders of magnitude, but their predictions must be revalidated on selected candidates [27].

Surrogate	Strength	Limitation
Local linear response	Interpretable; directly gives J_y and SVD modes	Valid only near a working point.
Polynomial / PCE	Fast and structured; useful for smooth maps	Complexity grows rapidly with dimension and order.
Gaussian process	Provides uncertainty for BO; kernels encode correlations	Exact training scales cubically with history size.
Neural network	Very fast inference; flexible high-dimensional map	Requires data, validation, and an uncertainty model.
Physics prior model	Injects first-order, high-order, or differentiable physics into BO	Can bias the search if model discrepancy is ignored.

Table B.2: Surrogate models for inverse beam matching. The surrogate approximates $\mathbf{k} \mapsto \mathbf{y}$ or $\mathbf{k} \mapsto \Phi$; the optimizer determines how the prediction and uncertainty are used.

B.3.2.3 Kernel design

The kernel in Eq. (B.3.13) is the main structural assumption of a GP. It determines which changes in machine parameters are considered similar and therefore how information is transferred between sampled points. The kernel should be chosen after considering the expected regularity, periodicity, separability, correlations, and fidelity of the beam-dynamics response [25, 29, 40]. In GPyTorch and Xopt, a base kernel is commonly wrapped by `ScaleKernel` to learn an output scale; automatic relevance determination (ARD) is activated through separate length scales for each input dimension; and `active_dims` can restrict a component kernel to a selected subset of controls [41, 40].

Table B.3: Principal kernel families relevant to UH beam-dynamics optimization. The table includes standard GPyTorch objects and physics-informed extensions that can be passed to Xopt through a custom covariance module.

Family	Principle and representative object	Appropriate UH use	Main caution
RBF / squared exponential	Infinitely differentiable stationary covariance; `RBFKernel`. ARD gives one length scale per control.	Very smooth matrix optics, differentiable low-order models, and well-behaved local scans.	Often too smooth across sharp aperture, loss, or branch boundaries.
Matérn	Finite differentiability controlled by ν; `MaternKernel` with $\nu = 1/2, 3/2, 5/2$.	Robust default for realistic accelerator objectives with moderate roughness or model discrepancy.	The selected ν imposes a fixed regularity class.
Rational quadratic	Scale mixture of RBF kernels; `RQKernel`.	Landscapes containing both broad and narrow response scales.	The scale-mixture interpretation can be weakly identifiable with little data.
Linear	Covariance proportional to $\mathbf{k}^\top\mathbf{k}'$; `LinearKernel`.	Local response regimes, residuals around a nominal optics model, and approximately affine calibration problems.	Cannot represent saturation, branches, or curved valleys by itself.

Table B.3 (continued)

Family	Principle and representative object	Appropriate UH use	Main caution
Polynomial	Covariance from powers of an inner product; `PolynomialKernel`.	Low-order nonlinear response when a Taylor-like structure is physically expected.	High powers can extrapolate poorly and become ill-conditioned.
Periodic / cosine	Correlation repeats with a learned period; `PeriodicKernel` or `CosineKernel`.	RF phase, phase-advance branches, or scans with genuine periodic structure.	Incorrect periodicity can create artificial copies of a solution.
Spectral mixture	Learns a mixture of spectral components; `SpectralMixtureKernel`.	Multiple repeat lengths, quasi-periodic responses, or data-informed frequency structure.	More hyperparameters and local likelihood optima; requires careful initialization.
Piecewise polynomial	Compactly supported or locally polynomial covariance; `PiecewisePolynomialKernel`.	Local response with weak long-range correlation or large histories where sparsity is useful.	Correlation can vanish too abruptly if the support is poorly selected.
Additive	Sum of kernels over groups or dimensions; `AdditiveKernel` or `AdditiveStructureKernel`.	Separating quadrupole, dipole, cavity, steering, or subsystem contributions.	Misses strong interactions unless cross-terms are added.
Product	Product of component kernels; `ProductKernel` or `ProductStructureKernel`.	Modeling interactions such as phase × amplitude or optics × energy dependence.	Products can become overly restrictive or numerically small in high dimension.
SVD/full-metric	Mahalanobis distance with a full matrix C, implemented by transformed inputs or a custom kernel.	Coupled machine knobs, Hessian/SVD modes, and rotated narrow valleys.	C must be regularized and updated when the local response changes.
Feature / deep	Apply a physics or neural feature map before a base kernel.	Correlation in optics, dispersion, phase, or learned latent coordinates rather than raw controls.	Uncertainty can be misleading outside the validity domain of the feature map.
Multitask / coregionalization	Joint covariance over inputs and outputs; `IndexKernel`, `Multitask Kernel`, or `LCMKernel`.	Correlated objectives and constraints, several diagnostics, or related beamline sections.	More data and identifiable cross-output structure are required.
Derivative-aware	Joint covariance of values and derivatives; `RBFKernelGrad`, `RBFKernelGradGrad`, or `Matern52KernelGrad`.	Incorporating Jacobians or automatic-differentiation gradients from a differentiable model.	Derivatives must use consistent normalization and trustworthy noise estimates.
Specialized domain	Examples include `Cylindrical Kernel`, discrete-index kernels, and custom mixed-variable kernels.	Bounded high-dimensional domains, categorical machine modes, or mixed continuous/discrete controls.	Use only when the domain geometry genuinely matches the kernel assumption.

For an axis-aligned ARD kernel, the distance metric is diagonal. UH beamline problems often require a rotated metric because magnet and RF controls act in correlated combinations. Let $V = [\mathbf{v}_1, \ldots, \mathbf{v}_n]$ be the right-singular-vector basis of J_y, a Hessian eigenbasis, or an active-subspace basis. Define

$$C = V\ \mathrm{diag}(\ell_1^2, \ldots, \ell_n^2)\, V^\top, \tag{B.3.21}$$

and use the Mahalanobis kernel

$$K_C(\mathbf{k}, \mathbf{k}') = \sigma_f^2 \exp\left[-\frac{1}{2}(\mathbf{k} - \mathbf{k}')^\top C^{-1} (\mathbf{k} - \mathbf{k}')\right]. \tag{B.3.22}$$

The off-diagonal entries of C encode correlations between physical controls. A useful initialization is to assign short modal length scales to large-response or high-curvature directions and longer scales to sloppy directions, for example

$$\ell_j \propto \left(\sigma_j^2 + \epsilon\right)^{-1/2} \qquad \text{or} \qquad \ell_j \propto \left(|\lambda_j(H_\Phi)| + \epsilon\right)^{-1/2}, \tag{B.3.23}$$

followed by marginal-likelihood training or validation. These expressions are initial physics scalings, not immutable rules: the correlation matrix should be regularized and allowed to adapt when data contradict the local response model.

A second option is a feature kernel,

$$K(\mathbf{k}, \mathbf{k}') = K_0\big(\mathcal{F}(\mathbf{k}), \mathcal{F}(\mathbf{k}')\big)\,, \tag{B.3.24}$$

where $\mathcal{F}$ may contain phase advances, beta beating, dispersion, response spectra, feasibility probabilities, or differentiable-model embeddings. This directly connects the kernel to the topology and physics features introduced in Sections B.1.4 and B.2.

B.3.2.4 Prior means

The mean $m(\mathbf{k})$ in Eq. (B.3.13) should not be treated as automatically zero or constant. A fast physics model, archived response model, or neural surrogate can supply

$$m(\mathbf{k}) = \Phi_{\text{prior}}(\mathbf{k}), \tag{B.3.25}$$

so that the GP learns the residual

$$\delta f(\mathbf{k}) = f(\mathbf{k}) - \Phi_{\text{prior}}(\mathbf{k}). \tag{B.3.26}$$

This is useful when an analytical, high-order, differentiable, or learned model captures broad trends but not high-fidelity corrections. A differentiable prior has the additional advantage that its gradients and local curvature can be reused to initialize response-based coordinates, derivative-aware kernels, or trust-region metrics, while the GP models the residual discrepancy and uncertainty. Prior-mean BO has been shown to reduce the number of accelerator evaluations when the prior is informative [32]. The prior must nevertheless be validated: an overconfident or systematically wrong mean can bias both exploitation and the apparent feasibility boundary. The residual uncertainty should therefore remain trainable, and low- versus high-fidelity disagreement should be stored as an explicit model-quality diagnostic.

B.3.3 Constraints and feasibility

Not all constraints should be treated in the same way. In the UH framework we distinguish forbidden conditions, probabilistic constraints, and soft penalties.

B.3.3.1 Forbidden conditions

These define the non-negotiable safe or valid set. Let $g_f^{(j)}(\mathbf{k}) \le 0$ denote forbidden conditions. Then

$$\mathcal{K}_{\text{safe}} = \Big\{\mathbf{k} \in \mathcal{K} : g_f^{(j)}(\mathbf{k}) \le 0,\ j = 1, \ldots, n_f\Big\}\,. \tag{B.3.27}$$

Points outside $\mathcal{K}_{\text{safe}}$ are rejected, masked, or assigned no utility. For UH, examples include beam loss, excessive beam size that violates aperture margin, and failure of the tracking or machine-evaluation procedure.

B.3.3.2 Probabilistic constraints

Some constraints are uncertain because of model discrepancy, shot noise, or incoming-beam jitter. In that case one imposes a chance constraint

$$\mathbb{P}\big[g_p^{(j)}(\mathbf{k}) \le 0\big] \ge 1 - \alpha_j, \tag{B.3.28}$$

where α_j is an acceptable risk level. In UH applications this category can include dispersion, emittance growth, and energy or energy-spread limits when uncertainty is appreciable and the same tolerance does not apply to every beamline section.

B.3.3.3 Soft penalties

Other requirements are preferences rather than exclusions. If $P_\ell(\mathbf{k}) \geq 0$ is a penalty term, minimize

$$\Phi_{\text{tot}}(\mathbf{k}) = \Phi_{\text{match}}(\mathbf{k}) + \sum_\ell w_\ell P_\ell(\mathbf{k}). \tag{B.3.29}$$

Typical examples include beta beating, phase-advance targets or bands, magnet-current smoothness or slew regularization, and in some studies unstable-transport indicators used as graded warnings rather than absolute vetoes.

Constraint class	Mathematical form	Typical UH examples	Optimizer implication
Forbidden	$g_f(\mathbf{k}) \leq 0$ with rejection outside $\mathcal{K}_{\text{safe}}$	beam loss; aperture/beam-size margin; tracking failure	hard feasibility filter; no candidate accepted outside the safe set.
Probabilistic	$\mathbb{P}[g_p(\mathbf{k}) \leq 0] \geq 1 - \alpha$	dispersion and D'_x when uncertain; emittance growth; energy or energy-spread limits	feasibility probability enters acquisition, ranking, or post-filtering.
Soft penalty	$\Phi_{\text{tot}} = \Phi + \sum w_\ell P_\ell$	beta beating; phase-advance bands; current or slew regularization; selected stability indicators	contributes to the scalar merit function and shapes the trade-off surface.

Table B.4: Constraint taxonomy used in the UH framework. The same physical quantity may move from soft to probabilistic or forbidden status depending on the machine section, operating mode, and risk tolerance.

B.3.3.4 Feasibility-aware acquisition

A deterministic solver such as SLSQP or `trust-constr` can handle explicit bounds and nonlinear constraints, but is not by itself aware of uncertainty. In BO, objective and constraint observables can be modeled separately. A simple constrained acquisition is

$$a_{\text{con}}(\mathbf{k}) = a_{\text{obj}}(\mathbf{k}) \prod_j P\big(g_p^{(j)}(\mathbf{k}) \leq 0\big), \tag{B.3.30}$$

combined with the hard mask $\mathbf{k} \in \mathcal{K}_{\text{safe}}$. This separates three logically distinct operations: reject forbidden points, weight uncertain feasibility probabilistically, and include preferences in the objective.

B.3.4 Local Bayesian optimization

A single stationary GP can become inefficient in a high-dimensional or multimodal hypercube because one model must describe several basins and response scales. Local Bayesian optimization fits probabilistic models inside one or several restricted regions and coordinates their allocation [31].

B.3.4.1 Trust regions

A trust region a with center $\mathbf{c}_a$ and side lengths $\boldsymbol{\Delta}_a$ may be written

$$\mathcal{T}_a = \{\mathbf{k} : |k_j - c_{a,j}| \leq \Delta_{a,j},\ j = 1, \ldots, n\}. \tag{B.3.31}$$

The local GP is trained primarily on points relevant to this neighborhood, reducing the burden of fitting one homogeneous model across unrelated topology branches.

B.3.4.2 Region adaptation

After successful improvements, the region expands; after repeated failures, it contracts. Abstractly,

$$\Delta_{a,j}^{(+)} = \eta_+ \Delta_{a,j}, \qquad \Delta_{a,j}^{(-)} = \eta_- \Delta_{a,j}, \qquad \eta_+ > 1 > \eta_- > 0. \tag{B.3.32}$$

Success and failure counters should be defined with respect to feasible improvement, not objective improvement alone. A point that reduces Φ while approaching a forbidden boundary should not automatically trigger expansion.

The initial shape should also reflect the local response. In the Hessian or SVD basis,

$$\Delta_{a,j} \propto \frac{1}{\sqrt{\lambda_{a,j}^{(H)} + \epsilon}} \tag{B.3.33}$$

produces shorter steps along stiff modes and longer steps along sloppy modes. The same metric can be inherited from the kernel correlation matrix C of Eq. (B.3.21) so that the local GP and trust region use compatible geometry.

B.3.4.3 Multiple regions

In TuRBO-m, several local regions operate simultaneously and a bandit-like rule allocates evaluations among them. For the UH topology-aware workflow, the number of initial regions should be connected to the pre-map,

$$N_{\mathrm{TR}} \approx \min\left(N_{\mathrm{basin}}^{\mathrm{good}}, N_{\mathrm{budget}}^{\mathrm{parallel}}, N_{\mathrm{model}}^{\mathrm{stable}}\right), \tag{B.3.34}$$

where the terms represent the number of promising basins, the parallel evaluation budget, and the number of regions that can each be supported by enough data. Distinct phase-advance branches or disconnected feasible islands should remain separate until evidence supports merging them.

B.3.4.4 Validity radius

A local linear model, Hessian model, prior-mean approximation, or GP is valid only over a finite neighborhood. Define

$$\rho_{\mathrm{val}}(\mathbf{k}_0) = \sup\left\{\rho : \epsilon_{\mathrm{pred}}(\rho) \le \epsilon_{\max}, \epsilon_{\mathrm{disc}}(\rho) \le d_{\max}, \sigma_{\mathrm{GP}}(\rho) \le \sigma_{\max}\right\}, \tag{B.3.35}$$

where ϵ_{pred} is validation error, ϵ_{disc} is low- versus high-fidelity discrepancy, and σ_{GP} is posterior uncertainty. Local derivative consistency can be added as a fourth criterion. The trust-region dimensions should not substantially exceed this validity radius. When the radius collapses, the model should be retrained, the region contracted, or a higher-fidelity evaluation requested.

B.3.5 Robust solutions

A low objective value is not enough. In many accelerator problems a numerically excellent point is operationally poor because it is too sensitive to perturbations in quadrupole gradients, dipole fields, RF amplitudes or phases, cavity geometry parameters, and other controls.

B.3.5.1 Parameter sensitivity

Near a candidate $\mathbf{k}^*$, sensitivity may be summarized by J_y, H_Φ, finite-difference perturbation ensembles, or uncertainty propagation. Large singular values or Hessian eigenvalues identify directions where small control errors strongly modify the observables or objective. A robustness score should therefore be evaluated in normalized engineering units and, where possible, using realistic calibration and jitter distributions rather than an arbitrary Euclidean perturbation.

B.3.5.2 Feasible neighborhoods

Let $M \succ 0$ be a metric derived from normalized controls, the SVD basis, or a Hessian. Define

$$\mathcal{N}_{\rho,\delta}(\mathbf{k}^*) = \{\mathbf{k} : \|\mathbf{k} - \mathbf{k}^*\|_M \le \rho,\ \Phi(\mathbf{k}) \le \Phi(\mathbf{k}^*) + \delta,\ \mathbf{k} \in \mathcal{K}_{\text{safe}}\}\,. \tag{B.3.36}$$

Large connected neighborhoods are operationally preferable to isolated, needle-like optima. The neighborhood volume, minimum distance to a forbidden boundary, and worst-case or probabilistic degradation under perturbations are complementary stability measures.

B.3.5.3 Solution families

The optimizer should retain clusters of acceptable solutions rather than only the single best point. Several independently discovered candidates may represent: (i) samples from one connected robust valley; (ii) distinct phase-advance branches with similar performance; or (iii) genuinely disconnected feasible basins. Clustering in the augmented feature space of Section B.2, followed by local connectivity and perturbation tests, distinguishes these cases. A family of nearby solutions is valuable for commissioning, recovery after machine drift, and selecting an operating point with additional unmodeled margin.

B.3.6 UH workflow

The preceding sections define the optimizer components. The UH workflow specifies how the topology map, Bayesian models, constraints, trust regions, and robustness tests are coordinated.

B.3.6.1 Topology-informed initialization

The pre-map supplies initial seeds, modal coordinates, local metrics, feasibility labels, and candidate basin centers. For a basin a, write

$$\mathbf{k} = \mathbf{k}_a + V_a\mathbf{q}, \tag{B.3.37}$$

where V_a is a local Hessian/SVD basis. A deterministic trust metric is

$$\|\Delta\mathbf{k}\|^2_{H_a} = \Delta\mathbf{k}^\top (H_a + \epsilon I)\Delta\mathbf{k}, \tag{B.3.38}$$

and an island-model covariance may be initialized as

$$C_a = \alpha_a (H_a + \epsilon I)^{-1}. \tag{B.3.39}$$

For BO, the same basin history can initialize a prior mean, a full-metric kernel, and one local trust region. This gives the deterministic, population-based, and Bayesian methods a common geometry rather than three unrelated initializations.

B.3.6.2 Adaptive sequence

A practical sequence for the Diagnostic Chicane, an accelerating linac, or a related UH system is:

1. Define the control domain, target observables, model fidelities, and the three constraint classes.

2. Generate a Sobol or other low-discrepancy pre-map and store objective, optics, phase, feasibility, and stability features.

3. Compute Jacobians and Gauss–Newton or exact Hessians on selected points. Store singular spectra and model-discrepancy tests.

4. Classify candidate basins and feasibility boundaries in augmented feature space.
5. Use Bayesian exploration to refine sparse regions, uncertain boundaries, and incomplete basins.
6. Select the surrogate mean, kernel family, correlation metric, and noise model using the physics and validation information available.
7. Launch deterministic refiners, BO, TuRBO, or island-model methods from representative basins.
8. Adapt trust regions according to feasible improvement and local validity radius.
9. Evaluate robustness neighborhoods and retain families of acceptable solutions.
10. Validate finalists with the highest-fidelity model and, when appropriate, experimental diagnostics.

The resulting logic is

$$\boxed{\begin{aligned}&\text{topology pre-map} + \text{ physics and feasibility labels}\\ &\quad\longrightarrow \text{ Bayesian exploration} \longrightarrow \text{ local or global exploitation}\\ &\quad\longrightarrow \text{ validity and robustness validation.}\end{aligned}} \tag{B.3.40}$$

The coordinated transition from topology pre-mapping to Bayesian exploration and local trust-region exploitation is illustrated in Figure B.1.

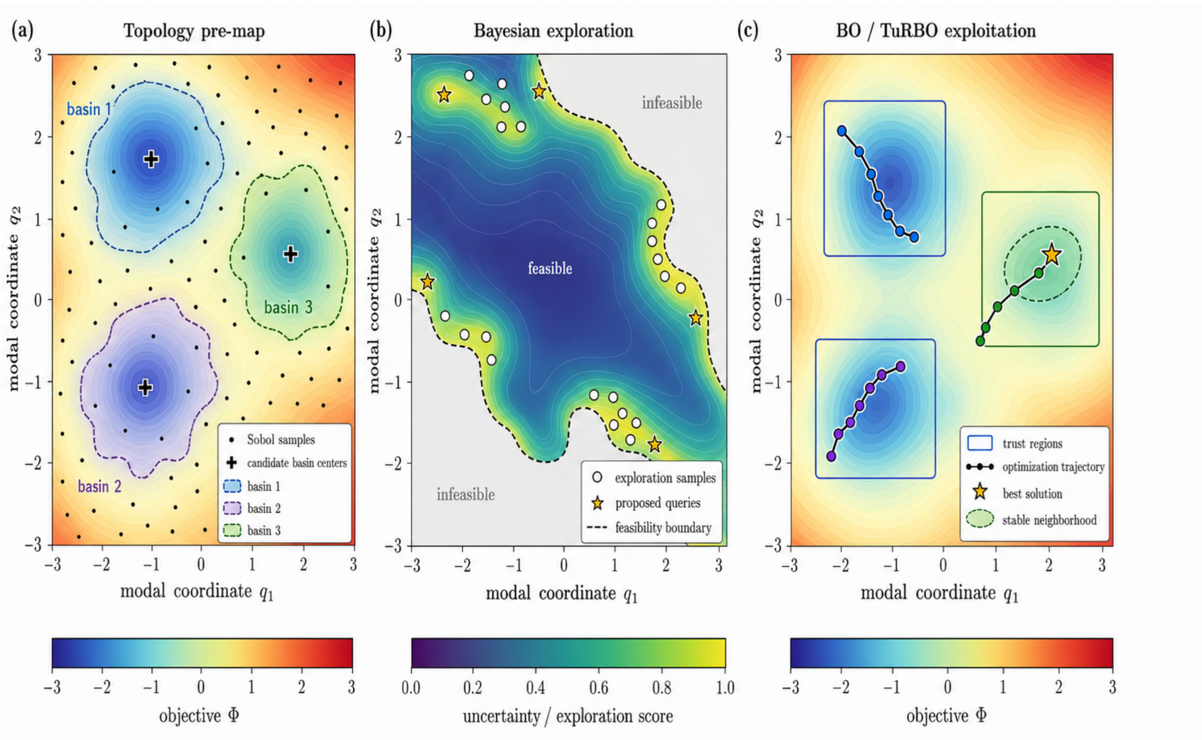

Figure B.1: Schematic of drafted UH optimization workflow in a reduced two-dimensional modal coordinate space (q_1, q_2). **(a)** A Sobol pre-map samples the admissible parameter domain and identifies and classifies basins. **(b)** Bayesian exploration preferentially samples uncertain regions and the boundaries of the predicted feasible domain. **(c)** Bayesian optimization or TuRBO initializes local trust regions around promising basins.

B.3.6.3 Problem regimes

Problem regime	Model information	Recommended mapping	Recommended optimizer
2–4 lattice knobs	First-order map; cheap derivatives	dense grid or Sobol; J_y and phase labels	SVD/pseudoinverse, SLSQP, Nelder–Mead benchmark.
5–20 quadrupoles in a beamline	first- or high-order map; derivatives available	tiered Sobol map; Hessian/SVD labels; graph basins	staged SLSQP/`trust-constr`; BO/TuRBO initialized by basins.
Many controls with expensive tracking	derivatives partial or expensive	low-discrepancy map plus surrogate; derivatives on a subset; boundary exploration	TuRBO, CMA-ES/islands, NN-assisted local refinement.
Machine/online FEL tuning	noisy black-box measurements; limited shots	historical scans; priors; feasibility and safety labels	GP BO/UCB/EI, prior-mean BO, safe exploration, robust local refinement.
RF gun / linac start-to-end	nonlinear collective effects; limited exact derivatives	sparse high-fidelity set; differentiable or NN/GP/PCE surrogate	multiobjective search, BO, active learning, and high-fidelity verification.

Table B.5: Possible UH optimization regimes. The recommended model and optimizer depend on dimensionality, derivative availability, evaluation cost, noise, and the severity of physical constraints.

Annex A: Linear Algebra for Response Maps

Let $A \in \mathbb{R}^{m\times n}$. Its eigenvalues are defined only for square matrices by

$$A\mathbf{v} = \lambda\mathbf{v}, \qquad \det(A - \lambda I) = 0, \tag{B.4.1}$$

and each eigenvector belongs to the null space

$$\mathbf{v} \in \ker(A - \lambda I). \tag{B.4.2}$$

For a non-square response matrix $J_y \in \mathbb{R}^{m\times n}$, the SVD is the correct modal decomposition:

$$J_y = U\Sigma V^\top, \qquad J_y v_j = \sigma_j u_j. \tag{B.4.3}$$

The right singular vectors v_j are orthogonal magnet-current modes; the left singular vectors u_j are orthogonal observable modes; and the singular values σ_j are local gains. The rank is the number of nonzero singular values,

$$r = \mathrm{rank}(J_y) = \#\{j : \sigma_j > 0\}. \tag{B.4.4}$$

The fundamental subspaces are

$$\mathrm{range}(J_y) = \mathrm{span}\{u_1, \ldots, u_r\}, \tag{B.4.5}$$
$$\ker(J_y) = \mathrm{span}\{v_{r+1}, \ldots, v_n\}, \tag{B.4.6}$$
$$\mathrm{range}(J_y^\top) = \mathrm{span}\{v_1, \ldots, v_r\}, \tag{B.4.7}$$
$$\ker(J_y^\top) = \mathrm{span}\{u_{r+1}, \ldots, u_m\}. \tag{B.4.8}$$

The Moore–Penrose pseudoinverse is

$$J_y^+ = V\Sigma^+U^\top, \qquad \Sigma^+ = \mathrm{diag}(1/\sigma_1, \ldots, 1/\sigma_r), \tag{B.4.9}$$

possibly with truncation or Tikhonov filtering for small σ_j. If $m < n$, then at least $n - r$ magnet modes are locally unconstrained by the chosen observables; if $m > n$, then at least $m - r$ observable combinations are locally unreachable.

Annex B: Local Geometry of the Optimization Landscape

Let $\Phi : \mathbb{R}^n \to \mathbb{R}$ be a smooth objective and let

$$\mathcal{L}_c = \{\mathbf{k} : \Phi(\mathbf{k}) = c\} \tag{B.5.1}$$

be a level set. If $\nabla\Phi(\mathbf{k}_0) \neq 0$, the tangent hyperplane to $\mathcal{L}_c$ at $\mathbf{k}_0$ is

$$T_{\mathbf{k}_0}\mathcal{L}_c = \{\delta\mathbf{k} : \nabla\Phi(\mathbf{k}_0)^\top \delta\mathbf{k} = 0\}. \tag{B.5.2}$$

At a critical point $\nabla\Phi(\mathbf{k}^*) = 0$, the Hessian controls the local geometry:

$$\Phi(\mathbf{k}^* + \delta\mathbf{k}) = \Phi(\mathbf{k}^*) + \frac{1}{2}\delta\mathbf{k}^\top H_\Phi(\mathbf{k}^*)\delta\mathbf{k} + O(\|\delta\mathbf{k}\|^3). \tag{B.5.3}$$

The eigenvalue signs classify the point: positive definite Hessian means a strict local minimum; negative definite means a local maximum; mixed signs indicate a saddle; zero eigenvalues indicate degeneracy or the need for higher-order terms. The Morse index

$$\mathrm{ind}(H_\Phi) = \#\{j : \lambda_j(H_\Phi) < 0\} \tag{B.5.4}$$

counts independent downhill directions at a critical point. In topology mapping, approximate critical points and saddle regions are inferred from small gradient norm, Hessian sign structure, graph descent, and persistence under changes in sampling resolution.

Annex C: Implementation Notes for Optimization and BO

This annex names the principal software objects and their roles without turning the framework into a code tutorial. The entries are deliberately unnumbered so that the annex remains a practical reference rather than an additional hierarchy of theoretical sections.

- **Layered software architecture.** A practical implementation contains five cooperating layers:
  - a forward-model layer based on the selected UH matrix, high-order, Xsuite, RF-Track, or higher-fidelity tracking model;
  - a derivative and machine-learning layer using finite differences, differential algebra, PyTorch automatic differentiation, Cheetah-compatible backpropagation, SVDs, Hessians, and model-discrepancy diagnostics;
  - a topology and data layer storing Sobol histories, basin labels, feasibility information, and stability features;
  - an optimization layer based on SciPy, Xopt, or direct BoTorch/GPyTorch models;
  - an orchestration and safety layer for approved offline and online workflows.

  The forward-model callable should return $\mathbf{y}(\mathbf{k})$, $\Phi(\mathbf{k})$, constraint observables, phase advances, dispersion, and optional robustness quantities.

- **Xopt objects.** Xopt provides the most direct accelerator-oriented interface [28, 29, 41]. The principal objects are:
  - `VOCS`, defining variables, objectives, constraints, observables, and constants;
  - `Evaluator`, wrapping the simulation or machine evaluation;
  - `Xopt`, coupling the generator, evaluator, and accumulated history;

- `BayesianExplorationGenerator`, for response mapping and uncertainty reduction;
- `ExpectedImprovementGenerator`, for improvement-based BO;
- `UpperConfidenceBoundGenerator`, for UCB-based BO;
- `MOBOGenerator`, for multiobjective optimization;
- `OptimizeTurboController`, for adaptive local optimization;
- `SafetyTurboController`, for safety-aware local search;
- `StandardModelConstructor`, with `mean_modules` and `covar_modules` for prior means and custom kernels;
- `max_travel_distances`, for limiting candidate motion in selected coordinates.

Forbidden conditions belong in hard bounds or constraint filters; probabilistic constraints require modeled observables and feasibility probabilities; soft penalties remain part of the scalar objective.

- **BoTorch and GPyTorch objects.** Direct BoTorch/GPyTorch use is appropriate when the Xopt abstraction is too restrictive [30, 40]. Relevant objects include:
  - models: `SingleTaskGP`, `ModelListGP`, and `SingleTaskMultiFidelityGP`;
  - `qExpectedImprovement` and `qNoisyExpectedImprovement`;
  - `qUpperConfidenceBound`;
  - `qNoisyExpectedHypervolumeImprovement`;
  - acquisition optimization through `optimize_acqf`.

  GPyTorch supplies mean modules and the kernel families summarized in Table B.3. Custom models are the natural route for full correlation matrices, derivative observations, feature kernels, residual learning, or joint objective-constraint models.

- **Cheetah, backpropagation, and PyTorch ML tools.** For UH, the main added value of Cheetah is not its basic first-order transport, which can already be supplied by existing UH models, but its differentiable execution path and direct integration with the PyTorch machine-learning ecosystem [36, 34, 35]. Cheetah lattice and beam parameters are represented by PyTorch tensors, and a loss constructed from the output of `Segment.track` can remain connected to the input machine parameters through the automatic-differentiation graph. The most useful tools are:
  - `requires_grad=True` and `torch.nn.Parameter`, to designate magnet, RF, geometry, calibration, or latent-model parameters to be differentiated;
  - `Tensor.backward()` and `torch.autograd.grad`, to backpropagate a scalar objective or selected observable through the beam simulation;
  - `torch.func.grad`, `jacrev`, and `jacfwd`, to obtain gradients and full Jacobians of beam observables with respect to machine parameters;
  - `torch.func.hessian`, `jvp`, and `vjp`, to compute curvature, Jacobian-vector products, and vector-Jacobian products without always assembling dense derivative tensors;
  - `torch.func.vmap`, to vectorize compatible parameter scans, derivative evaluations, ensemble models, or batched sensitivity calculations;
  - `torch.optim.Adam` and `torch.optim.LBFGS`, for gradient-based fitting, system identification, differentiable matching, or initialization of a later BO stage;

  - `torch.nn.Module`, to combine the differentiable beam model with neural correction models, learned calibration maps, residual surrogates, or differentiable control policies.

  These tools can support gradient-informed pre-mapping, automatic Jacobian and Hessian construction, system identification from diagnostic data, physics-informed neural networks, residual learning between model fidelities, and derivative-enhanced GP models. Their role in the UH framework is to enrich the derivative, feature, and prior-model layers; final candidates remain subject to the selected high-fidelity simulation and constraint checks.

- **Osprey orchestration.** Osprey is a supervisory and safety layer rather than a BO algorithm [38]. Offline, it can launch approved studies, update kernel, acquisition, or trust-region hyperparameters inside predefined ranges, and maintain an auditable record. Online, it can connect the approved Xopt/BoTorch capability to facility channels while enforcing channel boundaries and human approval for hardware writes. Thus Osprey may apply approved hyperparameter changes automatically, while beam-affecting writes remain subject to the chosen operator and machine-protection policy.

- **Stored data products.** For reproducibility, the pre-map and adaptive optimization database should store:

  - control vectors in engineering and normalized units;
  - raw observables, residuals, objectives, and individual penalty terms;
  - forbidden flags, probabilistic-constraint observables, and feasibility probabilities;
  - phase advances, dispersion, beta beating, singular values, Hessian spectra, condition numbers, and basin labels;
  - surrogate predictions, uncertainty, prior-mean values, trust-region identifiers, validity radii, and robustness scores;
  - random or Sobol seeds, model version, lattice file, incoming-beam definition, and all optimizer hyperparameters.

  This database is the persistent memory used to build topology maps, compare model fidelities, recover previous operating regions, and audit online decisions.

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